सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड
(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)
(भारत सरकार का उद्यम)
CENTRAL TRANSMISSION UTILITY OF INDIA LTD.
(A wholly owned subsidiary of Power Grid Corporation of India Limited)
(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/34 **Date:** 08-10-2024

As per distribution list

Subject: 34th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 34th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 20th September 2024 (Friday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

आशिष
08/10/24

(Kashish Bhambhani)
General Manager (CTU)

"सौदामिनी", प्रथम तल, प्लॉट सं.2, सेक्टर-29, गुरुग्राम- 122001 (हरियाणा), दूरभाष: 0124-2822000, सीआईएन: U40100HR2020GOI091857
"Saudamini", 1st Floor, Plot No. 2, Sector-29, Gurugram-122001 (Haryana), Tel.: 0124-2822000, CIN: U40100HR2020GOI091857
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Minutes of 34th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 20.09.2024

The 34th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 20/09/2024. List of participants is enclosed at **Annexure-XI**

It was informed that minutes of the 32nd CMETS-NR meeting held on 10.07.2024 was issued vide letter dated 17.07.2024. No comments were received, accordingly, minutes is confirmed as circulated.

It was informed that Minutes of the 33rd CMETS-NR meeting held on 05.08.2024 was issued vide letter dated 16.08.2024. No comments were received on the minutes, however, following corrections may be noted:

(i). At Page No. 23, for connectivity application of M/s Adani Renewable(2200000661), start date of connectivity was inadvertently mentioned as 31.12.2030. The same may be corrected and read as 31.12.2028 as mentioned in earlier para of the minutes for the same application.
(ii). At Page No. 42, for connectivity of M/s Renew Solar (2200000779), dedicated transmission line was mentioned as 400 kV S/c line. The same may be corrected and read as 400 kV S/c line on D/c tower.
(iii). At Page No. 49, for connectivity of M/s Adani Renewable(2200000810), dedicated transmission line was mentioned as 400 kV S/c line. The same may be corrected and read as 400 kV S/c line on D/c tower.
(iv). At Annexure- XI at Page No. 97, "*Additional system for connectivity at 220kV level only of Sanchore PS*" shall be deleted as it is already covered in element no. 2 of "*Transmission system for Connectivity under GNA at Sanchore PS*" mentioned above in same page.

With the above corrections, minutes are confirmed.

A. Transition of 675 MW (200+350+125) Connectivity Granted to Soltown Infra at Bikaner-II PS

It was informed that Soltown Infra Private Limited was granted cumulative connectivity of 675 MW (App. No. 12000003889 – 200 MW, 0212100007 – 350 MW & 0212100008 – 125 MW) at Bikaner-II PS through 220 kV D/c line under Connectivity Regulations 2009. CTUIL vide letter dated 05.04.2023 had revoked the entire 675 MW connectivity granted to M/s Soltown Infra Private Limited in light of blacklisting of M/s Soltown Infra from obtaining any connectivity or open access from CTU.

Subsequently, M/s Soltown Infra filed petition with CERC for setting aside the blacklisting letter issued by CTUIL and quash the revocation letter dated 05.04.2023. In the hearing held on 21.08.2023, the CERC directed the CTUIL not to allocate the revoked capacity (675 MW) at Bikaner- II S/s to any other entity till the next date of hearing. Subsequently, the interim protection was continued by the Commission till the issuance of Order.

CERC vide order dated 30.04.2024 on Petition No. 114/MP/2023 along with IA No. 28/2023 and 51/2023 set aside the CTUIL letter dated 05.04.2023 vide which the 675 MW Connectivity granted to SIPL was revoked & directed CTUIL to allow the M/s Soltown Infra to convert the 675 MW Connectivity, initially granted under the Connectivity Regulations, 2009, to Connectivity in compliance with the

Central Electricity Regulatory Commission (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022.

In view of the above order, CTUIL asked M/s Soltown Infra to opt/surrender of connectivity as pert 37.2 of GNA Regulations 2022 and submit the necessary documents required for transition. Subsequently, M/s Soltown infra opted to convert the 675 MW connectivity under GNA Regulations vide letter dated 20.08.2024 along with all necessary documents required for transition of connectivity. In view of the above, transition of 675 MW connectivity granted to Soltown Infra was taken up in the 34th CMETS NR meeting for deliberations. As only Stage-II connectivity was applied & granted to M/s Soltown Infra, transition of connectivity granted to M/s Soltown shall be treated under clause 37.2 of Connectivity & GNA Regulations 2022.

It was also informed that , CTUIL has filed a review petition 20 of 2024 in Petition No. 114/MP/2023 with CERC. Therefore, transition of connectivity granted to M/s Soltown under GNA Regulations shall be subject to outcome of the above review petition. M/s Soltown noted the same.

Further, M/s Soltown vide letter dated 06.09.2024 informed that due to unforeseen challenges, including land acquisition issues, their solar park now comprises two separate land patches, one which can accommodate 350 MW and another of 325 MW. Consequently, in all cases two distinct transmission lines will be required, each originating from the respective land patches and running separately to the PGCIL substation. Therefore, due to location constraints, instead of connectivity for the entire solar park through 220 kV D/c line as granted earlier, M/s Soltown requested for Connectivity of 350 MW portion (App. No. 0212100007) through a 220 kV S/c line suitable to carry 350 MW. For the remaining two applications of 125 MW (App. No. 0212100008) & 200 MW (App. No. 1200003889) they have requested connectivity through a separated 220 kV S/c line suitable to carry 325 MW(125+200).

CTU enquired status of land submitted by M/s Soltown at the application stage. M/s Soltown clarified that earlier land remains in their possession. Soltown infra also clarified that due to land acquisition issues their project land is not contiguous and therefore they would like to go with two separate S/c lines from each of their project locations. CTUIL informed that the implementing common pooling station has many advantages including economics of scale like installation of S/s equipments, reactive compensation devices like SVG, filters etc. Due to separation of pooling stations for 350 MW & 325 MW, Soltown infra may be required to install separate reactive compensation devices, filters etc. at each generation pooling station. M/s Soltown infra agreed for the same and informed that they shall not request for any change of DTL configuration in future after this. After deliberations, it was agreed to revise the DTL configuration from D/c line to two separate S/c lines for 350 MW & 325 MW. Further, while granting connectivity, reserved capacity of 675 MW as per CERC order was considered.

Accordingly, details of transition of Connectivity granted to M/s Soltown Infra under 37.2 is as below:

S. No.	Applicant Name	Application No.(St-II)	Pooling Station	Connectivity (MW)	Bay Scope	Start Date of Connectivity as per Stage-II intimation	Start Date requested for transition under GNA	Start Date of Connectivity under GNA	Conn BG Requirement
1.	Soltown Infra Private Limited	1200003889	Bikaner-II PS	**200**	Applicant	30-06-2025	31-12-2025	31-12-2025	• Conn BG-1: Rs. 50 lakhs • Conn BG-2: Nil (Bay under applicant scope) • Conn BG-3: Rs. 4 Cr.

It was informed that M/s Soltown has requested for Start date of connectivity from 31.12.2025 for the above application. Considering the same & expected commissioning date of the transmission system (i.e. 27.12.2025), the start date of Connectivity under GNA shall be 31.12.2025 as requested.

M/s Soltown infra vide mail dated 30.09.2024 confirmed that the tower configuration for their S/c line shall be Single circuit line on double circuit tower.

Accordingly, it was agreed to transition connectivity of 200 MW granted to M/s Soltown Infra at Bikaner-II PS under GNA Regulations 2022. Details of transmission system for Connectivity under GNA is as below:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

(i). Common Pooling Station of Soltown Infra Private Limited (0212100008-125 MW & 1200003889-200 MW) RE Power Parks – Bikaner-II PS 220 kV S/c line on D/c tower(Suitable to carry minimum 325 MW at nominal voltage).

C. Transmission system for Connectivity under GNA:

1) Augmentation of 1x500MVA (8th) 400/220 kV ICT at Bikaner –II PS
2) Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG)
3) Augmentation of 2x1500 MVA (3rd & 4th)765/400 kV ICTs at Bikaner-III Pooling Station
4) Augmentation of 2x1500 MVA (3rd & 4th),765/400 kV ICTs at Neemrana-II S/s
5) LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS
6) Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad)
7) Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVAr switchable line reactor for each circuit at each end
8) Neemrana-II -Kotputli 400 kV D/c line (Quad)
9) LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s
10) Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAr switchable line reactor for each circuit at each end
11) Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAr switchable line reactor for each circuit at each end
12) Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG)

Start Date of Connectivity under GNA: 31-12-2025

S. No.	Applicant Name	Application No.(St-II)	Pooling Station	Connectivity (MW)	Bay Scope	Start Date of Connectivity as per Stage-II intimation	Start Date requested for transition under GNA	Start Date of Connectivity under GNA	Conn BG Requirement
2.	Soltown Infra Private Limited	0212100007	Bikaner-II PS	**350**	Applicant	30-06-2025	31-10-2025	22.08.2026	• Conn BG-1: Rs. 50 lakhs • Conn BG-2: Nil (Bay under applicant scope) • Conn BG-3: Rs. 7 Cr.

It was informed that M/s Soltown has requested for Start date of connectivity from 31.10.2025 for the above application. However, considering the same & expected commissioning of the transmission system (i.e. 22.08.2026), the start date of Connectivity under GNA shall be 22.08.2026.

M/s Soltown infra vide mail dated 30.09.2024 confirmed that the tower configuration for their S/c line shall be Single circuit line on double circuit tower.

It was also informed that, as per the application priority, the NR-WR inter regional corridor is also required for grant of connectivity under GNA Regulations 2022 for above application.

Accordingly, it was agreed to transition connectivity of 350 MW granted to M/s Soltown Infra at Bikaner-II PS under GNA Regulations 2022. Details of transmission system for Connectivity under GNA is as below:

A. <u>Associated Transmission System (ATS):</u> NIL

B. <u>Transmission System under applicant scope</u>

(i). Soltown Infra Private Limited RE Power Park – Bikaner-II PS 220 kV S/c line on D/c tower (Suitable to carry minimum 350 MW at nominal voltage).

C. Transmission system for Connectivity under GNA:

1) Augmentation of 1x500MVA (9th) 400/220 kV ICT at Bikaner –II PS
2) Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG)
3) Augmentation of 2x1500 MVA (3rd & 4th)765/400 kV ICTs at Bikaner-III Pooling Station
4) Augmentation of 2x1500 MVA (3rd & 4th),765/400 kV ICTs at Neemrana-II S/s
5) LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS
6) Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad)
7) Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVAr switchable line reactor for each circuit at each end
8) Neemrana-II -Kotputli 400 kV D/c line (Quad)
9) LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s
10) Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAr switchable line reactor for each circuit at each end
11) Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAr switchable line reactor for each circuit at each end
12) Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG)

S. No.	Applicant Name	Application No.(St-II)	Pooling Station	Connectivity (MW)	Bay Scope	Start Date of Connectivity as per Stage-II intimation	Start Date requested for transition under GNA	Start Date of Connectivity under GNA	Conn BG Requirement

Additional Transmission system for inter-regional power transfer:

1) Establishment of 765 kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors
2) Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at each end and
3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line

OR

3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAr 765 kV bus reactor
3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAr switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line
3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar

Start Date of Connectivity under GNA: 22-08-2026

S. No.	Applicant Name	Application No.(St-II)	Pooling Station	Connectivity (MW)	Bay Scope	Start Date of Connectivity as per Stage-II intimation	Start Date requested for transition under GNA	Start Date of Connectivity under GNA	Conn BG Requirement
3.	Soltown Infra Private Limited	0212100008	Bikaner-II PS	**125**	Applicant	30-06-2025	31-12-2025	22.08.2026	• Conn BG-1: Rs. 50 lakhs • Conn BG-2: Nil (Bay under applicant scope) • Conn BG-3: Rs. 2.5Cr.

It was informed that M/s Soltown has requested for Start date of connectivity from 31.12.2025 for the above application. However, considering the same & expected commissioning of the transmission system (i.e. 22.08.2026), the start date of Connectivity under GNA shall be 22.08.2026.

It was also informed that, as per the application priority, the NR-WR inter regional corridor is also required for grant of connectivity under GNA Regulations 2022 for above application.

Accordingly, it was agreed to transition connectivity of 125 MW granted to M/s Soltown Infra at Bikaner-II PS under GNA Regulations 2022. Details of transmission system for Connectivity under GNA is as below:

A. **Associated Transmission System (ATS):** **NIL**

S. No.	Applicant Name	Application No.(St-II)	Pooling Station	Connectivity (MW)	Bay Scope	Start Date of Connectivity as per Stage-II intimation	Start Date requested for transition under GNA	Start Date of Connectivity under GNA	Conn BG Requirement

B. Transmission System under applicant scope

(i). Common Pooling Station of Soltown Infra Private Limited (0212100008-125 MW & 1200003889-200 MW) RE Power Parks – Bikaner-II PS 220 kV S/c line on D/c tower (Suitable to carry minimum 325 MW at nominal voltage).

C. Transmission system for Connectivity under GNA:

1) Augmentation of 1x500MVA (9th) 400/220 kV ICT at Bikaner–II PS
2) Augmentation of 1x1500 MVA (4th) 765/400kV ICT at Bikaner (PG)
3) Augmentation of 2x1500 MVA (3rd & 4th)765/400 kV ICTs at Bikaner-III Pooling Station
4) Augmentation of 2x1500 MVA (3rd & 4th),765/400 kV ICTs at Neemrana-II S/s
5) LILO of both ckts of 400 kV Bikaner(PG)-Bikaner-II D/c line (Quad) at Bikaner-III PS
6) Bikaner-II PS – Bikaner-III PS 400 kV D/c line (Quad)
7) Bikaner-III - Neemrana-II 765 kV D/c line along with 330 MVAr switchable line reactor for each circuit at each end
8) Neemrana-II -Kotputli 400 kV D/c line (Quad)
9) LILO of both ckts of 400 kV Gurgaon (PG) - Sohna Road (GPTL) D/c line (Quad) at Neemrana-II S/s
10) Bikaner-III - Neemrana-II 765 kV D/c line (2nd) along with 330 MVAr switchable line reactor for each circuit at each end
11) Neemrana-II- Bareilly (PG) 765 kV D/c line along with 330 MVAr switchable line reactor for each circuit at each end
12) Augmentation by 400/220 kV, 1x500 MVA ICT (3rd) at Kotputli (PG)

Additional Transmission system for inter-regional power transfer:

1) Establishment of 765 kV Mandsaur Pooling Station along with 2x330 MVAR (765 kV) Bus Reactors
2) Beawar- Mandsaur PS 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at each end and
3a) Mandsaur PS – Indore(PG) 765 kV D/c Line along with 1x330 MVAR switchable line reactor on each ckt at Mandsaur end of Mandsaur PS – Indore(PG) 765 kV D/c Line

OR

3b) Establishment of 765 kV Kurawar S/s with 2x330 MVAr 765 kV bus reactor
3c) Mandsaur – Kurawar 765 kV D/c line along with 240 MVAr switchable line reactors on each ckt at both ends of Mandsaur – Kurawar 765 kV D/c line
3d) LILO of Indore – Bhopal 765 kV S/c line at Kurawar

Start Date of Connectivity under GNA: 22-08-2026

It was informed that as per 37.2(d), in case the entity exercises the option to convert the Connectivity granted under the Connectivity Regulations, 2009 as Connectivity under these Regulations in terms of option (i) of clause (a) of this regulation, the Nodal Agency shall, within next 30 days, intimate the amount of Conn-BG1, Conn-BG2 and Conn-BG3, to be paid by such entity in terms of Regulation 8 of these regulations, after adjusting bank guarantee, if any, paid by such entity under the Connectivity Regulations, 2009.

As per 37.2(e) of GNA Regulations, entity need to submit applicable Conn BGs within 2 months of intimation by CTUIL.

As per 37.2(f), On furnishing of Conn-BG1, Conn-BG2 and Conn-BG3 under clause (e) of this Regulation, existing agreements between the entity and the Nodal Agency shall be aligned with provisions of Regulation 10.3 of these regulations.

As per 37.2(g) On alignment of existing agreements under clause (f) of this Regulation, the entity shall become Connectivity grantee for all purposes under these regulations.

Further, as per 37.2(h),
" In case the entity fails to furnish Conn-BG1, Conn-BG2 and ConnBG3 as intimated by the Nodal Agency in terms of clause (d) of this Regulation,
(i) Connectivity granted to the entity under the Connectivity Regulations, 2009 shall be revoked by the Nodal Agency and
(ii) Conn-BG1 furnished, if any, under the Connectivity Regulations, 2009 shall be encashed and
(iii) Conn-BG2 furnished, if any under the Connectivity Regulations, 2009, shall be returned:
Provided that in case the construction of terminal bay has been awarded for implementation under ISTS through CTU, Conn-BG2 furnished under the Connectivity Regulations, 2009 shall be encashed "

In view of the above, M/s Soltown need to submit the Conn BGs mentioned above within the stipulated timeline as per GNA Regulations 2022. Further, in the meeting, it was mentioned that transition of connectivity granted to M/s Soltown under GNA Regulations shall be subject to outcome of the review petition filed by CTUIL. Subsequently, CERC vide order dated 30.09.2024 has disposed of the Review petition No. 20/RP/2024 by CTUIL.

B. Application related matters in Northern Region

B1. RE Applications for Connectivity deferred in the 33rd CMETS NR meeting held on 05.08.2024 (Applications received in May'24):

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity Location (As per Application)	Conn BGs requirement
1.	2200000788	ACME Solar Holdings Limited (erstwhile ACME Solar Holdings	Sirohi distt., Rajasthan	07.05.2024	Generator (Solar)	Land BG Route	300	31.03.2027	Sirohi PS	NA

<table>
<tr><th>Sl. No.</th><th>Application ID</th><th>Applicant</th><th>Project Location</th><th>Submission Date</th><th>Nature of Applicant</th><th>Criterion for applying</th><th>Connectivity Quantum (MW)</th><th>Start Date of Connectivity (As per Application)</th><th>Connectivity Location (As per Application)</th><th>Conn BGs requirement</th></tr>
<tr><td></td><td></td><td>Private Limited)</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="11">It was informed that M/s ACME vide mail dated 16.09.2024 has withdrawn the above application. Accordingly, the above application shall be closed as per GNA Regulations.</td></tr>
<tr><td>2.</td><td>2200000793</td><td>SAEL Industries Limited</td><td>Sirohi distt., Rajasthan</td><td>10.05.2024</td><td>Generator (Solar)</td><td>Land BG Route</td><td>300</td><td>30.05.2026</td><td>Sirohi PS</td><td>NA</td></tr>
<tr><td colspan="11">It was informed that M/s SAEL Industries vide mail dated 17.09.2024 has withdrawn their application. Accordingly, the above application shall be closed as per GNA Regulations.</td></tr>
<tr><td>3.</td><td>2200000794</td><td>SAEL Industries Limited</td><td>Sirohi distt., Rajasthan</td><td>10.05.2024</td><td>Generator (Solar)</td><td>Land BG Route</td><td>300</td><td>30.05.2026</td><td>Sirohi PS</td><td>NA</td></tr>
<tr><td colspan="11">It was informed that M/s SAEL Industries vide mail dated 17.09.2024 has withdrawn their application. Accordingly, the above application shall be closed as per GNA Regulations.</td></tr>
<tr><td>4.</td><td>2200000827</td><td>Amplus Centaur Solar Private Limited</td><td>Sirohi distt., Rajasthan</td><td>20.05.2024</td><td>Generator (Solar)</td><td>Land BG Route</td><td>350</td><td>30.04.2027</td><td>Sirohi PS</td><td>• Conn BG1: Rs. 50 Lakh
• Conn BG2: Rs. 3 Cr.
• Conn BG3: Rs. 2 lakh/MW</td></tr>
<tr><td colspan="11">In the 33rd CMETS-NR meeting, it was informed that, in view of grant of 2100 MW connectivity at Sirohi PS, above application by M/s Amplus Centaur shall be taken up for discussion for grant in the next CMETS NR meeting after identification of additional transmission system required for evacuation of power beyond 2100 MW from Sirohi PS. It may be noted that transmission system for evacuation of power from Sirohi was planned for net evacuation capacity of 2000 MW only.

From the studies, considering upcoming RE generation in Northern and Western region by 2028-29 timeframe, severe transmission constraints were observed in Western region under N-1 contingency. Therefore, considering the severe evacuation constraints towards Western Region from Sirohi complex as well as low Short Circuit Ratio at 220kV level of Sirohi PS, it was proposed to grant connectivity to subsequent applications in Sirohi complex at proposed Sirohi (HVDC) PS (Sirohi-II) for which evacuation shall be through proposed Sirohi HVDC scheme.

Accordingly, it was proposed to grant connectivity to M/s Amplus Centaur at 220 kV Sirohi (HVDC) PS (Sirohi-II) through S/c line. M/s Amplus Centaur enquired about any possibility of getting connectivity at Sirohi PS. CTUIL informed that as explained earlier, due to constraints in Western Region, there is no margin available at Sirohi PS, therefore connectivity is being granted at Sirohi (HVDC) PS (Sirohi-II). M/s Amplus Centaur enquired if they can get 7 days time to decide on their application. Regarding this CTUIL informed that, soon they shall issue minutes and intimation. M/s Amplus Centaur in the meantime may take a decision and inform the same to CTUIL. In case no communication is received from M/s Amplus Centaur for withdrawal of connectivity, the intimation shall be issued as agreed in this CMETS meeting. M/s Amplus agreed for the same.</td></tr>
</table>

<table>
<tr><th>Sl. No.</th><th>Application ID</th><th>Applicant</th><th>Project Location</th><th>Submission Date</th><th>Nature of Applicant</th><th>Criterion for applying</th><th>Connectivity Quantum (MW)</th><th>Start Date of Connectivity (As per Application)</th><th>Connectivity Location (As per Application)</th><th>Conn BGs requirement</th></tr>
<tr><td colspan="11">Applicant was asked to confirm the 220 kV bay scope at Sirohi (HVDC) PS (Sirohi-II) end and DTL tower configuration. M/s Amplus Centaur informed that the 220 kV bay shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that, the transmission system for evacuation of power from Sirohi, Jalore & Sanchore complexes through Sirohi(HVDC) is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Amplus Centaur Solar Private Limited has requested Start Date of Connectivity under GNA from 30.04.2027. However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim). M/s Amplus Centaur noted the same.

Accordingly, it was agreed to grant 350 MW connectivity to M/s Amplus Centaur Solar Private Limited at 220 kV Sirohi(HVDC) PS. Details of Transmission system for connectivity of M/s Amplus Centaur Solar Private Limited under GNA is as below:

<u>Details of Transmission system for Connectivity under GNA:</u>

A. <u>Associated Transmission System (ATS):</u> NIL

B. <u>Transmission System under applicant scope</u>

(i). M/s Amplus Centaur Solar Private Limited Solar Power Project– Sirohi (HVDC) PS (Sirohi-II) 220 kV S/c line on D/c tower (suitable to carry minimum 350 MW at nominal voltage)

C. <u>Transmission system for Connectivity under GNA:</u> As per Annexure-I

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system</td></tr>
<tr><td>5.</td><td>2200000841</td><td>ACME Solar Holdings Limited (erstwhile ACME Solar Holdings Private Limited)</td><td>Jodhpur distt., Rajasthan</td><td>31.05.2024</td><td>Generator (Solar)</td><td>Land BG Route</td><td>100</td><td>31.03.2027</td><td>Nagaur (MERTA-II) PS</td><td>• Conn BG1: Rs. 50 Lakh
• Conn BG2: Nil(In sharing)
• Conn BG3: Rs. 2 lakh/MW</td></tr>
<tr><td colspan="11">It was informed that M/s ACME vide mail dated 16.09.2024 has withdrawn the above application. Accordingly, the above application shall be closed as per GNA Regulations.</td></tr>
<tr><td>6.</td><td>2200000891</td><td>Serentica Renewables India Private Limited</td><td>Udaipur distt.,</td><td>31.05.2024</td><td>Generator (Solar)</td><td>Land BG Route</td><td>400</td><td>30.06.2026</td><td>Rishabhdeo PS</td><td>NA</td></tr>
</table>

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity Location (As per Application)	Conn BGs requirement
			Rajasthan							

It was informed that M/s Serentica vide mail dated 19.09.2024 has withdrawn the above application. Accordingly, the above application shall be closed as per GNA Regulations.

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity Location (As per Application)	Conn BGs requirement
7.	2200000860	Enren-III Energy Private Limited	Nagaur distt., Rajasthan	31.05.2024	Generator (Solar)	Land BG Route	300	31.01.2027	Nagaur Complex (Merta-II) PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

In the 33rd CMETS-NR meeting, it was informed that in view of grant of 2100 MW connectivity at Merta-II PS, the above applications by M/s Enren-III Energy shall be taken up for discussion for grant in the next CMETS NR meeting after identification of additional transmission system required for evacuation of power beyond 2100 MW from Merta-II PS. It may be noted that transmission system for evacuation of power from Merta-II PS was planned for net evacuation capacity of 2000 MW only.

From the studies, considering upcoming RE generation in Northern and Western region by 2028-29 timeframe, severe transmission constraints were observed in Western region under N-1 contingency. Therefore, considering the severe evacuation constraints in Western Region as well as low Short Circuit Ratio of Merta-II PS, it was proposed to grant connectivity to subsequent applications in Merta/Nagaur complex at proposed Merta-III PS for which evacuation shall be through proposed Beawar HVDC scheme.

Accordingly, it was proposed to grant connectivity to M/s Enren-III Energy at 220 kV Merta-III PS through S/c line. M/s Enren-III Energy agreed for the same. Applicant was asked to confirm the 220 kV bay scope at Merta-III PS end and DTL tower configuration. M/s Enren-III Energy informed that the 220 kV bay shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that, the transmission system for evacuation of power from Nagaur(Merta) & Pali complexes through Beawer(HVDC) is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Enren-III Energy Private Limited has requested Start Date of Connectivity under GNA from 31.01.2027, However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2031), the start date of Connectivity under GNA shall be 31.03.2031 (interim). M/s Enren-III Energy noted the same.

Accordingly, it was agreed to grant 300 MW connectivity to M/s Enren-III Energy Private Limited at 220 kV Merta-III PS. Details of Transmission system for connectivity of M/s Enren-III Energy Private Limited under GNA is as below :

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity Location (As per Application)	Conn BGs requirement
B. <u>Transmission System under applicant scope</u> (i). M/s Enren-III Energy Private Limited Solar Power Project – Merta-III PS 220 kV S/c line on D/c tower(suitable to carry minimum 300 MW at nominal voltage) **C. <u>Transmission system for Connectivity under GNA:</u>** As per **Annexure-II** **Start Date of Connectivity under GNA: 31.03.2031(Interim).** Final date shall be confirmed upon award of the system										
8.	2200000894	Sindhari Solar Power Private Limited	Nagaur distt., Rajasthan	31.05.2024	Renewable Power Park developer (Solar)	Land BG Route	300	30.06.2027	Nagaur	• Conn BG1: Rs. 50 Lakh • Conn BG2: As per bay scope • Conn BG3: Rs. 2 lakh/MW
M/s Sindhari Solar in the meeting informed that the timeline of Merta-III PS is not matching with their generation project timeline expected in 2027. Accordingly, they shall withdraw their application. CTUIL asked M/s Sindhari Solar to confirm the same in mail. Subsequently, M/s Sindhari Solar vide mail dated 24.09.2024 confirmed their decision to withdraw the application. Accordingly, it was decided to close the above application as per the GNA Regulations.										

A2. RE Applications for Connectivity deferred in the 33rd CMETS NR meeting held on 05.08.2024 (Applications received in June'24):

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity Location (As per Application)	Conn BGs requirement
1.	2200000832	Prespa Solar Private Limited	Nagaur distt., Rajasthan	04.06.2024	Renewable Power Park developer (Solar)	Land BG Route	250	31.12.2026	Nagaur Complex (Merta-II PS)	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW
In the 33rd CMETS-NR meeting, it was informed that in view of grant of 2100 MW connectivity at Merta-II PS, the above application by M/s Prespa Solar shall be taken up for discussion for grant in the next CMETS NR meeting after identification of additional transmission system required for evacuation of power beyond 2100 MW from Merta-II PS. It may be noted that transmission system for evacuation of power from Merta-II PS was planned for net evacuation capacity of 2000 MW only.										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity Location (As per Application)	Conn BGs requirement

From the studies, considering upcoming RE generation in Northern and Western region by 2028-29 timeframe, severe transmission constraints were observed in Western region under N-1 contingency. Therefore, considering the severe evacuation constraints in Western Region as well as low Short Circuit Ratio of Merta-II PS, it was proposed to grant connectivity to subsequent applications in Merta/Nagaur complex at proposed Merta-III PS for which evacuation shall be through proposed Beawer HVDC scheme.

Accordingly, it was proposed to grant connectivity to M/s Prespa Solar at 220 kV Merta-III PS through S/c line. M/s Prespa Solar agreed for the same.

Applicant was asked to confirm the 220 kV bay scope at Merta-III PS end and DTL tower configuration. M/s Prespa Solar informed that the 220 kV bay shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that, the transmission system for evacuation of power from Nagaur(Merta) & Pali complexes through Beawar(HVDC) is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that, M/s Prespa Solar Private Limited has requested Start Date of Connectivity under GNA from 31.12.2026, However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2031), the start date of Connectivity under GNA shall be 31.03.2031 (interim). M/s Prespa Solar noted the same.

Accordingly, it was agreed to grant 250 MW connectivity to M/s Prespa Solar Private Limited at 220 kV Merta-III PS. Details of Transmission system for connectivity of M/s Prespa Solar Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). M/s Prespa Solar Private Limited Renewable Power Park – Merta-III PS 220 kV S/c line on D/c tower(suitable to carry minimum 300 MW at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-II**

Start Date of Connectivity under GNA: 31.03.2031(Interim). Final date shall be confirmed upon award of the system

A3. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 received in July'24:

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
1.	2200001021	Sunsure Solarpark RJ One Private Limited	Bikaner distt., Rajasthan	05.07.2024	Generator (Solar)	Land BG Route	350	31.03.2029	Bikaner-V PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Sunsure Solarpark RJ One Private Limited has applied for connectivity for 350 MW at Bikaner-V PS. Accordingly, it was proposed to grant connectivity at Bikaner-V PS at 220 kV level through 1 no. of bay. M/s Sunsure Solarpark agreed for the same. Applicant was asked to confirm the 220 kV bay scope at Bikaner-V PS end and the DTL tower configuration. M/s Sunsure informed that the 220 kV bay shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that, the transmission system for evacuation of power from Bikaner-V PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Sunsure Solarpark RJ One Private Limited has requested Start Date of Connectivity under GNA from 31.03.2029. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2030), the start date of Connectivity under GNA shall be 31.03.2030 (interim). M/s Sunsure noted the same.

Accordingly, it was agreed to grant 350 MW connectivity to M/s Sunsure Solarpark RJ One Private Limited at 220 kV Bikaner-V PS. Details of Transmission system for connectivity of M/s Sunsure Solarpark RJ One Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

(i). M/s Sunsure Solarpark RJ One Private Limited Solar Power Project – Bikaner-V PS 220 kV S/c line on D/c tower (suitable to carry minimum 350 MW at nominal voltage)

C. Transmission system for Connectivity under GNA: As per **Annexure-III**

Start Date of Connectivity under GNA: 31.03.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
2.	2200000983	Sunbreeze Renewables Seven Private Limited	Ramgarh distt., Rajasthan	05.07.2024	Renewable Power Park developer (Solar)	Land Route	400	31.03.2026	Ramgarh-Jaisalmer	• Conn BG1: Rs. 50 Lakh • Conn BG2: Nil(Bay under Applicant Scope) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Sunbreeze Renewables Seven Private Limited has applied for connectivity for 400 MW at Ramgarh PS. However, Ramgarh PS is already closed for further grant. Further, Ramgarh-II PS was also closed for further grant (except enhancement available upto 100 MW in 220 kV bay granted to RJS Renewable (150 MW)) in 33rd CMETS NR meeting held on 05.08.2024.

In the 33rd CMETS-NR meeting it was mentioned that, with total connectivity of 5010 MW granted at Ramgarh-II PS & considering additional 884 MW capacity granted at Ramgarh PS & being evacuated through Ramgarh-II HVDC system, total connectivity granted with Ramgarh-II HVDC system shall be about 5894 MW. Further, there is margin available at 220 kV bay allocated to M/s RJS Renewables (150 MW). As Ramgarh-II HVDC is planned for 6000 MW, Ramgarh-II PS shall be closed for further grant of connectivity except enhancements (upto 100 MW in 220 kV bay allocated to RJS Renewable). Accordingly, subsequent applications in Ramgarh complex shall be considered for grant at subsequent pooling station (Ramgarh-III PS) in Ramgarh complex for which system is yet to be evolved.

Recently M/s Juniper Green Energy Private Limited (App No. 2200000756:360MW) which was agreed for grant at Ramgarh-II PS in 33rd CMETS NR meeting has withdrawn their connectivity application. However, even after considering the withdrawal of M/s Juniper, sufficient margin is not available for grant of 400 MW connectivity at Ramgarh-II PS. Further, the applicant being an Renewable Power Park developer cannot opt to reduce the connectivity quantum as it is mandated to obtain connectivity equal to the authorized quantum by State/Central government as applicable. Accordingly, it was proposed to grant connectivity Ramgarh-III PS at 220 kV level through 1 no. of bay.

M/s Sunbreeze Renewables requested CTUIL whether they can be accommodated connectivity at Ramgarh-II PS as 360 MW connectivity is now withdrawn and additional 100 MW margin is still available. CTUIL informed that in case 400 MW connectivity is granted at Ramgarh-II PS through a separate bay, then the enhancement margin available in RJS Renewable (150 MW) bay is to be reduced from 100 MW to 60 MW. Therefore, total connectivity of 210 MW(150+60) can only be accommodated at RJS Renewable bay. However, in case of any margin being created at Ramgarh-II PS due to withdrawal/closure, as Conn-BGs are yet to be submitted by applicants at Ramgarh-II PS, the margin in this bay may be replenished. After deliberations with stakeholders, it was decided that, considering that the Ramgarh-II substation is already closed for further grant, connectivity may be granted to M/s Sunbreeze Renewable at 220kV Ramgarh-II PS through a S/c line. Further, the enhancement margin available in RJS Renewable Bay shall be reduced to 60 MW.

Applicant was asked to confirm the 220 kV bay scope at Ramgarh-II PS end and the tower configuration. M/s Sunbreeze Renewable informed that the 220 kV bay shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. Subsequently, M/s Sunbreeze Renewable vide mail dated 27.09.2024 informed that they would like to keep 220 kV bay at Ramgarh-II PS end under the applicant scope. The same is noted.

It was mentioned that, the transmission system for evacuation of power from Ramgarh-II PS is presently under evolution. Upon evolution, the scheme shall

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that, M/s Sunbreeze Renewables Seven Private Limited has requested Start Date of Connectivity under GNA from 31.03.2026. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2030), the start date of Connectivity under GNA shall be 31.03.2030 (interim). M/s Sunbreeze Renewables noted the same.

Accordingly, it was agreed to grant 400 MW connectivity to M/s Sunbreeze Renewables Seven Private Limited at 220 kV Ramgarh-II PS. Details of Transmission system for connectivity of M/s Sunbreeze Renewables Seven Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

(i). M/s Sunbreeze Renewables Seven Private Limited Renewable Power Park – Ramgarh-II PS 220 kV S/c line on D/c tower (suitable to carry minimum 400 MW at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-IV**

Start Date of Connectivity under GNA: 31.03.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
3.	2200001056	Juniper Green Energy Private Limited	Bikaner distt., Rajasthan	16.07.2024	Generator (Solar)	Land BG Route	400	30.06.2029	Bikaner-IV PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Juniper Green Energy Private Limited has applied for 400 MW connectivity at Bikaner-IV PS. However, there is no margin available at Bikaner-IV PS. Accordingly, it was proposed to grant connectivity at Bikaner-V PS at 220 kV level through 1 no. of bay. M/s Juniper Green Energy agreed

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

for the same. Applicant was asked to confirm the 220 kV bay scope at Bikaner-V PS end and the DTL tower configuration. M/s Juniper Green Energy informed that the 220 kV bay at Bikaner-V PS end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that the transmission system for evacuation of power from Bikaner-V PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that, M/s Juniper Green Energy Private Limited has requested Start Date of Connectivity under GNA from 30.06.2029. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2030), the start date of Connectivity under GNA shall be 31.03.2030 (interim). M/s Juniper Green Energy noted the same.

Accordingly, it was agreed to grant 400 MW connectivity to M/s Juniper Green Energy Private Limited at 220 kV Bikaner-V PS. Details of Transmission system for connectivity of M/s Juniper Green Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). M/s Juniper Green Energy Private Limited Solar Power Project – Bikaner-V PS 220 kV S/c line on D/c tower (suitable to carry minimum 400 MW at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-III**

Start Date of Connectivity under GNA: 31.03.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
4.	2200001057	Foxtrot Solar Renewables Energy Private Limited	Barmer distt., Rajasthan	17.07.2024	Generator (Solar)	Land Route	100	31.12.2030	Barmer II PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

It was informed that M/s Foxtrot Solar Renewables Energy Private Limited has applied for connectivity for 100 MW at Barmer-II PS. However, there is no margin available at Barmer-II PS (except enhancement of 50 MW). Further, another application for connectivity of 80 MW by Foxtrot Solar Renewables Energy Private Limited (App. No. 2200001088) at Barmer-II PS is also being taken up for discussion in this meeting as per application priority. Accordingly, it was proposed to grant connectivity at Barmer-III PS at 220 kV level through 220 kV S/c line for the combined capacity of 180 MW. M/s Foxtrot Solar agreed for the same. Applicant was asked to confirm the 220kV bay scope at Barmer-III PS & DTL tower configuration. M/s Foxtrot Solar informed that the 220 kV bay at Barmer-III PS end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that, the transmission system for evacuation of power from Barmer-III PS is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that, M/s Foxtrot Solar Renewables Energy Private Limited has requested for Start Date of Connectivity under GNA from 31.12.2030. Considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 31.12.2030 (interim) as requested. M/s Foxtrot Solar noted the same.

Accordingly, it was agreed to grant 100 MW connectivity to M/s Foxtrot Solar Renewables at 220 kV Barmer-III PS. Details of Transmission system for connectivity of M/s Foxtrot Solar Renewables Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). Common Pooling station for M/s Foxtrot Solar Renewables Energy Private Limited (App. No. 2200001057 (100 MW) & 2200001088 (80MW) Solar Power Project – Barmer-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW per circuit at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-V**

Start Date of Connectivity under GNA: 31.12.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
5.	2200001065	ACME Cleantech Solutions Private Limited	Jaisalmer distt., Rajasthan	17.07.2024	Generator (Solar)	Land BG Route	150	31.10.2026	Fatehgarh-II PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Nil (Bay In sharing) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s ACME Cleantech Solutions Private Limited has applied for connectivity of 150 MW at Fatehgarh-II PS. M/s ACME Cleantech is already granted connectivity of 850 MW at Fatehgarh-II PS through 400 kV S/c line for its 2 nos. of applications (App. No. 2200000387(600 MW) 2200000396(250 MW)).

It was also informed that Fatehgarh-II PS was earlier closed with total connectivity grant of 5460 MW. However, 1000 MW of connectivity was surrendered during the GNA transition process in 2023. Therefore, the margin of 1000 MW was available for grant at Fatehgarh-II PS. With 850 MW connectivity already granted to M/s ACME, remaining margin at 400 kV Fatehgarh-II PS is 150 MW. Accordingly, it was proposed to grant connectivity for the above application at Fatehgarh-II PS through same 400 kV S/c line in sharing with earlier granted applications of M/s ACME Cleantech Solutions Pvt Ltd. (App. No. 2200000387(600 MW) & 2200000396(250 MW)). M/s ACME agreed for the same.

Further, it was mentioned that M/s ACME Cleantech Solutions Pvt Ltd. has requested Start Date of Connectivity under GNA from 31.10.2026. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2027), the start date of Connectivity under GNA shall be 31.03.2027 (interim). M/s ACME Cleantech noted the same.

Accordingly, it was agreed to grant 150 MW connectivity to M/s ACME Cleantech Solutions Private Limited at 400 kV Fatehgarh-II PS. Details of Transmission system for connectivity of ACME Cleantech Solutions Pvt Ltd. under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

(i). Common Pooling Station for Acme Cleantech Solutions Pvt Ltd. (App. No. 2200000387(600 MW), 2200000396(250 MW) & 22000001065(150MW)) Solar Power Projects– Fatehgarh-II PS 400 kV S/c line on D/c tower (suitable to carry minimum 1000 MW per circuit at nominal voltage)

C. Transmission system for Connectivity under GNA: As per Annexure-VI

Start Date of Connectivity under GNA: 31.03.2027(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
It was also informed that, with this grant of 150 MW to M/s ACME Cleantech, total connectivity granted at Fatehgarh-II PS shall be 5460 MW. As available margin at Fatehgarh-II PS is fully utilised, the Pooling station shall be closed for grant and no additional connectivity shall be granted at Fatehgarh-II PS. Any subsequent application received for grant at Fatehgarh-II PS shall be considered for grant in other RE pooling stations in Fatehgarh-Barmer complex as per application priority.										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
6.	2200001026	Renew Solar Power Private Limited	Barmer distt., Rajasthan	19.07.2024	Generator (Solar)	REMCL LOA	200	30.09.2026	Barmer PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW
It was informed that M/s Renew Solar Power Private Limited has applied for connectivity for 200 MW at Barmer PS. However, there is no margin available at Barmer-I PS (except enhancement of 50 MW) & Barmer-II PS (except enhancement of 50 MW). Accordingly, it was proposed to grant connectivity at 220 kV Barmer-III PS through 220 kV S/c line for the 200 MW capacity. M/s Renew agreed for the same. Applicant was asked to confirm the 220kV bay scope at Barmer-III PS end and DTL tower configuration. M/s Renew Solar informed that the 220 kV bay at Barmer-III PS end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted. It was mentioned that, the transmission system for evacuation of power from Barmer-III PS is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme. It was also informed that, M/s Renew Solar Power Private Limited has requested Start Date of Connectivity under GNA from 30.09.2026. Considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim). M/s Renew Solar noted the same. Accordingly, it was agreed to grant 200 MW connectivity to M/s Renew Solar Power Private Limited at 220 kV Barmer-III PS. Details of Transmission system for connectivity of M/s Renew Solar Power Private Limited under GNA is as below: Transmission system for connectivity of Renew Solar Power Private Limited under GNA is as below: **<u>Details of Transmission system for Connectivity under GNA:</u>**										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

(i). Renew Solar Power Private Limited Solar Power Project – Barmer-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW per circuit at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-V**

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
7.	2200001044	Adani Renewable Energy Holding Eighteen Limited	Phalodi distt., Rajasthan	24.07.2024	Generator (Solar)	Land Route	150 (Reduced to 50 MW)	15.06.2025	Bhadla - II PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Nil (In sharing) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Adani Renewable Energy Holding Eighteen Limited has applied for connectivity of 150 MW at Bhadla-II PS. However, there is no margin available for grant of connectivity of 150 MW at Bhadla-II PS (except enhancement of 50 MW) & Bhadla-III PS (except enhancement of 50 MW). M/s Adani Renewable Energy Holding Four Limited (App No. 2200000811) was earlier granted connectivity of 765 MW at of Bhadla-IV PS through 400kV S/c line. Accordingly, it was proposed to grant connectivity of 150 MW to M/s Adani Renewable Energy Holding Eighteen Limited at 400 kV Bhadla-IV PS in sharing with M/s Adani Renewable Energy Holding Four Limited (App No. 2200000811:765MW) solar power project.

M/s Adani in the meeting informed that they would opt to reduce the connectivity quantum to 50 MW if they can get connectivity at Bhadla-II at 400 kV in sharing with Adani Renewable Energy Holding Four (500 MW).

Regarding the M/s Adani's request, it was deliberated that out of a total of 350 MW margin vacated at Bhadla-II, 300 MW is already allocated to M/s Green Infra. Therefore, total connectivity granted with Azure Bay is already 950 MW which is more than the requirement of 900 MW. As the bay is fully utilized there is no issue in granting connectivity of 50 MW to M/s Adani Renewable Energy Holding Eighteen at 400 kV bay allocated to M/s Adani Renewable Energy Holding Four Limited (500 MW), which is underutilized against its capacity of 900 MW. Accordingly, after discussion with stakeholders, it was agreed to grant connectivity of 50 MW to M/s Adani Renewable Energy Holding Eighteen Ltd. at Bhadla-II PS in sharing with 400 kV bay allocated to M/s Adani Renewable

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

Energy Holding Four Limited (500 MW). M/s Adani was asked to confirm the reduction of connectivity quantum in mail/letter and submit the revised CEA installed capacity certificate. M/s Adani agreed for the same. Subsequently, M/s Adani vide mail dated 24.09.2024 confirmed the request to reduce the connectivity quantum from 150 MW to 50 MW along with the revised CEA installed capacity certificate.

It was also informed that M/s Adani Renewable Energy Holding Eighteen Limited has requested Start Date of Connectivity under GNA from 15.06.2025. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2027), the start date of Connectivity under GNA shall be 31.03.2027 (interim). M/s Adani Renewable noted the same.

Details of Transmission system for connectivity of M/s Adani Renewable Energy Holding Eighteen Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). Through sharing of dedicated transmission system of M/s Adani Renewable Energy Holding Four Limited (App. No. 1200003685 (500 MW)) Solar Power Project – Bhadla-II PS 400 kV S/c line on D/c tower (suitable to carry minimum 900 MW at nominal voltage)

   (ii). Infrastructure required for sharing dedicated transmission system - under the scope of M/s Adani Renewable Energy Holding Eighteen Limited

C. **Transmission system for Connectivity under GNA:** As per **Annexure-X**

Start Date of Connectivity under GNA: 31.03.2027(Interim). Final date shall be confirmed upon award of the system

It was also informed that, with this grant of 50 MW to M/s Adani, the margin available at Bhadla-II PS is completely exhausted. Accordingly, Bhadla-II PS shall be closed for further grant including enhancements.

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
8.	2200001089	TEQ Green Power XV Private Limited	Pali distt., Rajasthan	26.07.2024	Renewable Power Park developer (Solar)	Land BG Route	300	15.06.2027	Pali PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s TEQ Green Power XV Private Limited has applied for connectivity of 300 MW at Pali PS. Accordingly, it was proposed to grant connectivity at Pali PS at 220 kV level through 220 kV S/c line. M/s TEQ Green agreed for the same. Applicant was asked to confirm the 220kV bay scope at Pali PS end & DTL tower configuration. M/s TEQ Green informed that the 220 kV bay at Pali PS end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that, the transmission system for evacuation of power from Nagaur(Merta) & Pali complexes through Beawar(HVDC) is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s TEQ Green Power XV Private Limited has requested Start Date of Connectivity under GNA from 15.06.2027. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2031), the start date of Connectivity under GNA shall be 31.03.2031 (interim). M/s TEQ Green noted the same.

Accordingly, it was agreed to grant 300 MW connectivity to M/s TEQ Green Power XV Private Limited at 220 kV Pali PS. Details of Transmission system for connectivity of M/s TEQ Green Power XV Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). M/s TEQ Green Power XV Private Limited Renewable Power Park – Pali PS 220 kV S/c line on D/c tower (suitable to carry minimum 300 MW at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-VIII**

Start Date of Connectivity under GNA: 31.03.2031(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
9.	2200001087	TEQ Green Power XV Private Limited	Jalore distt., Rajasthan	26.07.2024	Renewable Power Park developer (Solar)	Land BG Route	300	15.06.2027	Jalore PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s TEQ Green Power XV Private Limited has applied for connectivity of 300 MW at Jalore PS. Accordingly, it was proposed to grant connectivity at Jalore PS at 220 kV through 1 no. of 220 kV line bay. M/s TEQ Green agreed for the same. Applicant was asked to confirm the 220kV bay scope at Jalore PS end & DTL tower configuration. M/s TEQ Green informed that the 220 kV bay at Jalore PS end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that the transmission system for evacuation of power from Sirohi, Jalore & Sanchore complexes through Sirohi(HVDC) is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s TEQ Green Power XV Private Limited has requested Start Date of Connectivity under GNA from 15.06.2027. However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim). M/s TEQ Green noted the same.

Accordingly, it was agreed to grant 300 MW connectivity to M/s TEQ Green Power XV Private Limited at 220 kV Jalore PS. Details of Transmission system for connectivity of M/s TEQ Green Power XV Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). TEQ Green Power XV Private Limited Renewable Power Park – Jalore PS 220 kV S/c line on D/c tower (suitable to carry minimum 300 MW per circuit at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-IX**

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
10.	2200001088	Foxtrot Solar Renewables Energy Private Limited	Barmer distt., Rajasthan	26.07.2024	Generator (Solar)	Land Route	80	31.12.2031	Barmer II PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Nil(Bay In sharing) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Foxtrot Solar Renewables Energy Private Limited has applied for connectivity for 80 MW at Barmer-II PS. As decided with the previous application of M/s Foxtrot (App. No. 2200001057 – 100 MW), it was proposed to grant connectivity at Barmer-III PS at 220 kV level through same 220 kV S/c line allocated with App. No. 2200001057.

It was mentioned that the transmission system for evacuation of power from Barmer-III is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Foxtrot Solar Renewables Energy Private Limited has requested Start Date of Connectivity under GNA from 31.12.2031. Considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 31.12.2031 (interim) as requested. M/s Foxtrot Solar noted the same.

Accordingly, it was agreed to grant 80 MW connectivity to M/s Foxtrot Solar Renewables at 220 kV Barmer-III PS. Details of Transmission system for connectivity of Foxtrot Solar Renewables Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). Common Pooling station for M/s Foxtrot Solar Renewables Energy Private Limited (App. No. 2200001057 (100 MW) & 2200001088 (80MW) Solar Power Project – Barmer-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW per circuit at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-V**

Start Date of Connectivity under GNA: 31.12.2031(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
11.	2200001085	Avaada Energy Private Limited	Bikaner distt., Rajasthan	26.07.2024	Generator (Solar)	NHPC LOA	700	30.04.2026	Bhadla - III PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Avaada Energy Private Limited has applied for connectivity of 700 MW at Bhadla-III PS. However, there is no margin available for grant of connectivity of 700 MW at Bhadla-III PS. Accordingly, it was proposed to grant connectivity to M/s Avaada Energy at Bhadla-IV PS through 400 kV bay (1 no.). M/s Avaada Energy enquired whether they can be granted connectivity at 220 kV voltage level. Regarding this, CTUIL informed that M/s Avaada has applied single application for 700 MW connectivity quantum for which it is not optimal to grant connectivity at 220 kV as it would require additional 400/220 kV ICTs & incur additional losses in ISTS. Further, from techno economical point of view, 700 MW Quantum can be granted at 400 kV in ISTS. Accordingly, it was proposed to grant connectivity at 400 kV. M/s Avaada noted the same.

Applicant was asked to confirm the 400 kV bay scope at Bhadla-IV PS end and the tower configuration. M/s Avaada Energy informed that the 400 kV bay at Bhadla-IV PS end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that the transmission system for evacuation of power from Bhadla-IV is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Avaada Energy Private Limited has requested Start Date of Connectivity under GNA from 30.04.2026. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2030), the start date of Connectivity under GNA shall be 31.03.2030 (interim). M/s Avaada Energy noted the same.

Accordingly, it was agreed to grant 700 MW connectivity to M/s Avaada Energy at 400 kV Bhadla-IV PS. Details of Transmission system for connectivity of M/s Avaada Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). Avaada Energy Private Limited Solar Power Project – Bhadla-IV PS 400 kV S/c line on D/c tower (Suitable to carry minimum 900 MW at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-VII**

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
Start Date of Connectivity under GNA: 31.03.2030(Interim). Final date shall be confirmed upon award of the system										

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
12.	2200001100	Diyos Three Renewables Private Limited	Pali distt., Rajasthan	31.07.2024	Renewable Power Park developer (Solar)	Land BG Route	300	01.04.2031	Pali PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Diyos Three Renewables Private limited has applied for connectivity of 300 MW at Pali PS. Accordingly, it was proposed to grant connectivity at Pali PS at 220 kV level through 220 kV S/c line. M/s Diyos Three agreed for the same. Applicant was asked to confirm the 220kV bay scope at Pali PS & DTL tower configuration. M/s Diyos Three informed that the 220 kV bay at Pali PS end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that that the transmission system for evacuation of power from Nagaur (Merta) & Pali complexes through Beawar (HVDC) is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Diyos Three Renewables Private limited has requested Start Date of Connectivity under GNA from 01.04.2031. Considering the same and expected commissioning date of transmission system (i.e. 31.03.2031), the start date of Connectivity under GNA shall be 01.04.2031 (interim) as requested. M/s Diyos Three noted the same.

Accordingly, it was agreed to grant 300 MW connectivity to M/s Diyos Three Renewables Private Limited at 220 kV Pali PS. Details of Transmission system for connectivity of M/s Diyos Three Renewables Private limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Injection at ISTS point (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

(i). Diyos Three Renewables Private Limited Renewable Power Park – Pali PS 220 kV S/c line on D/c tower (suitable to carry minimum 300 MW at nominal voltage)

C. <u>Transmission system for Connectivity under GNA:</u> As per **Annexure-VIII**

Start Date of Connectivity under GNA: 01.04.2031(Interim). Final date shall be confirmed upon award of the system

B4. Application for Grant of GNARE to entities other than STU under Regulation 20.1, 20.3 and 20.4 to entities under Regulation 17.1(ii), (iii), (v) and (vi):

Sl No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA$_{RE}$ within Region (MW)	GNA$_{RE}$ outside Region (MW)	Total Quantum (MW) of GNA$_{RE}$ Required	Start date of GNA$_{RE}$	End Date of GNA$_{RE}$
1.	2200001060	Asian Fine Cements Private Limited	17.07.2024	NR	Drawee entity connected to Intra State Transmission System	0	4	4	01.09.2024	31.03.2026

It was informed that, Asian Fine Cements Private Limited has applied for GNA$_{RE}$ of 4 MW (Outside the region) as a drawee entity connected to Intra State Transmission System of Punjab (PSTCL) at 220 kV Rajpura S/s. In this regard, Asian Fine cements has submitted NOC from PSTCL along with the application.

It was mentioned that Punjab has been granted GNA of 5497 MW as per 18.1 of GNA regulation. Further, about 61 MW of GNA/GNA$_{RE}$ is granted to intra state connected entities of Punjab. Considering above, there is sufficient margin available in the existing ISTS system for drawl of additional 4 MW from ISTS periphery of Punjab (PSTCL) network.

In view of the above, it was agreed to grant 4 MW of GNA$_{RE}$ (outside the region) to Asian Fine Cements Private Limited for drawl of power from ISTS network. It was also informed that M/s Asian Fine has requested for Start Date of GNA$_{RE}$ from 01.09.2024. M/s Asian Fine Cements was asked to confirm the revised start date as the date has already elapsed. CTU also advised applicant to keep realistic start date of GNA to avoid such issues.

M/s Asian Fine Cements informed that they would like to revise their Start Date to 01.12.2024. CTUIL asked Asian Fine Cements to confirm the same in mail. Subsequently, M/s Asian Fine Cements vide mail dated 20.09.2024 confirmed their start date as 01.12.2024. Accordingly, it was agreed to grant GNA$_{RE}$ of 4 MW (Outside Region) to M/s Asian Fine Cements through existing system with Start date from 01.12.2024.

Sl No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNARE within Region (MW)	GNARE outside Region (MW)	Total Quantum (MW) of GNARE Required	Start date of GNARE	End Date of GNARE
2.	2200001058	Ambuja Cements Limited	17.07.2024	NR	Drawee entity connected to Intra State Transmission System	0	10	10	01.09.2024	31.03.2026

It was informed that Ambuja Cements Limited has applied for GNARE of 10 MW (Outside the region) as a drawee entity connected to Intra State Transmission System of PSTCL at 220 kV RTP Ropar S/s. In this regard, Ambuja cements Ltd. has submitted NOC from PSTCL along with the application.

It was mentioned that Punjab has been granted GNA of 5497 MW as per 18.1 of GNA regulation. Further, about 65 MW of GNA/GNARE is granted to intra state connected entities of Punjab. Considering above, there is sufficient margin available in the existing ISTS system for drawl of additional 10 MW from ISTS periphery of Punjab (PSTCL) network.

In view of the above, it was agreed to grant 10 MW of GNARE (outside the region) to Ambuja Cements Limited for drawl of power from ISTS network. M/s Ambuja Cements Limited has requested for Start Date of GNARE from 01.09.2024. However, as the requested start date has already passed, M/s Ambuja Cements Limited was asked to re confirm the start date as it is already elapsed. CTU also advised applicant to keep realistic start date of GNA to avoid such issues.

M/s Ambuja Cements informed that they would like to revise their Start Date to 01.12.2024. CTUIL asked Ambuja Cements to confirm the same in mail. Subsequently, M/s Ambuja Cements vide mail dated 20.09.2024 confirmed their start date as 01.12.2024. Accordingly, it was agreed to grant GNARE of 10 MW (Outside Region) to M/s Ambuja Cements through existing system with Start date from 01.12.2024.

Sl No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNARE within Region (MW)	GNARE outside Region (MW)	Total Quantum (MW) of GNARE Required	Start date of GNARE	End Date of GNARE
3.	2200001059	Ambuja Cements Limited	17.07.2024	NR	Drawee entity connected to Intra State Transmission System	0	3	3	01.09.2024	31.03.2026

It was informed that Ambuja Cements Limited has applied for GNARE of 3 MW (Outside the region) as a drawee entity connected to Intra State Transmission System of PSTCL at 66 kV Sub Station Malout Road (further connected to 220 kV Sub Station GNDTP Bathinda). In this regard, Ambuja Cements Limited has submitted NOC from PSTCL along with the application.

It was mentioned that Punjab has been granted GNA of 5497 MW as per 18.1 of GNA regulation. Further, about 75 MW of GNA/GNARE is granted to intra state connected entities of Punjab. Considering above, there is sufficient margin available in the existing ISTS system for drawl of additional 3 MW from ISTS periphery of Punjab (PSTCL) network.

Sl No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA$_{RE}$ within Region (MW)	GNA$_{RE}$ outside Region (MW)	Total Quantum (MW) of GNA$_{RE}$ Required	Start date of GNA$_{RE}$	End Date of GNA$_{RE}$

In view of the above, it was agreed to grant 3 MW of GNA$_{RE}$(outside the region) to Ambuja Cements Limited for drawl of power from ISTS network. M/s Ambuja Cements Limited has requested for Start Date of GNA$_{RE}$ from 01.09.2024. However, as the requested start date has already passed, M/s Ambuja Cements Limited was asked to re confirm the start date as it is already elapsed. CTU also advised applicant to keep realistic start date of GNA to avoid such issues.

M/s Ambuja Cements informed that they would like to revise their Start Date from 01.12.2024. CTUIL asked Ambuja Cements to confirm the same in mail. Subsequently, M/s Ambuja Cements vide mail dated 20.09.2024 confirmed their start date as 01.12.2024. Accordingly, it was agreed to grant GNA$_{RE}$ of 3 MW (Outside Region) to M/s Ambuja Cements through existing system with Start date from 01.12.2024

Sl No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA$_{RE}$ within Region (MW)	GNA$_{RE}$ outside Region (MW)	Total Quantum (MW) of GNA$_{RE}$ Required	Start date of GNA$_{RE}$	End Date of GNA$_{RE}$
4.	2200001081	ACC Limited	26.07.2024	NR	Drawee entity connected to Intra State Transmission System	0	4.5	4.5	01.09.2024	31.03.2026

It was informed that ACC Limited has applied for GNA$_{RE}$ of 4.5 MW (Outside the region) as a drawee entity connected to Intra State Transmission System of UPPTCL at 132 kV 132 kV S/s Gauriganj. In this regard, Ambuja Cements Limited has submitted NOC from UPPTCL from along with the application.

It was mentioned that Uttar Pradesh has been granted GNA of 10339 MW. Further, 100 MW of GNA is granted to Noida Power(NPCL) as a drawee entity connected to intra state connected entities of UP. Considering above, there is sufficient margin available in the existing ISTS system for drawl of additional 4.5 MW from ISTS periphery of Uttar Pradesh (UPPTCL) network.

In view of the above, it was agreed to grant 4.5 MW of GNA$_{RE}$ (outside the region) to ACC Limited for drawl of power from ISTS network. M/s ACC Limited has requested for Start Date of GNA$_{RE}$ from 01.09.2024. However, as the requested Start date has already passed, M/s ACC Limited was asked to re confirm the start date as it is already elapsed. CTU also advised applicant to keep realistic start date of GNA to avoid such issues.

M/s ACC Limited informed that they would like to revise their Start Date from 01.12.2024. CTUIL asked ACC Limited to confirm the same in mail. Subsequently, M/s ACC Limited vide mail dated 20.09.2024 confirmed their start date as 01.12.2024. Accordingly, it was agreed to grant GNA$_{RE}$ of 4.5 MW (Outside Region) to M/s ACC Limited through existing system with Start date from 01.12.2024

Sl No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA_{RE} within Region (MW)	GNA_{RE} outside Region (MW)	Total Quantum (MW) of GNA_{RE} Required	Start date of GNA_{RE}	End Date of GNA_{RE}
5.	2200001101	Ambuja Cements Limited	31.07.2024	NR	Drawee entity connected to Intra State Transmission System	0	5.85	5.85	01.09.2024	31.03.2026

It was informed that Ambuja Cements Limited has applied for GNA_{RE} of 5.85 MW (Outside the region) as a drawee entity connected to Intra State Transmission System of UPCL at 33/11 kV Raipur S/s of Discom, which is further connected to 132/33 kV S/s Bhagwanpur (PTCUL)). In this regard, Ambuja Cements Limited has submitted NOC from along with the application.

It was mentioned that Uttarakhand has been granted GNA of 1402 MW as per 18.1 of GNA regulation. Further, 8 MW of GNA_{RE} is granted to intra state connected entities of Uttarakhand (PTCUL). Considering above, there is sufficient margin available in the existing ISTS system for drawl of additional 5.85 MW from ISTS periphery of Uttarakhand (PTCUL) network.

In view of the above, it was agreed to grant 5.85 MW of GNA_{RE} (outside the region) to Ambuja Cements Limited for drawl of power from ISTS network. M/s Ambuja Cements Limited has requested for Start Date of GNA_{RE} from 01.09.2024. However, as the of requested Start date has already passed, M/s Ambuja Cements Limited was asked to re-confirm the start date as it is already elapsed. CTU also advised applicant to keep realistic start date of GNA to avoid such issues.

M/s Ambuja Cements informed that they would like to revise their Start Date from 01.12.2024. CTUIL asked Ambuja Cements to confirm the same in mail. Subsequently, M/s Ambuja Cements vide mail dated 20.09.2024 confirmed their start date as 01.12.2024. Accordingly, it was agreed to grant GNA_{RE} of 5.85 MW (Outside Region) to M/s Ambuja Cements through existing system with Start date from 01.12.2024

B. Network Expansion Scheme

B.1 Deletion of 400 kV Gopalpur and associated Transmission system for Connectivity/LTA/GNA

As part of Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase-II- Part G1, following transmission scheme was approved in 6th meeting of NCT held in 30.09.19.

- Removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh /Gopalpur S/s. Extension of above LILO section from Maharani Bagh/ Gopalpur upto Narela S/s so as to form Maharanibagh – Narela 400kV D/c(Quad) and Maharanibagh -Gopalpur-Narela 400kV D/c(Quad)lines
- 2 no of line bays at Narela each for Maharanibagh – Narela 400kV D/c (Quad) and Maharanibagh -Gopalpur-Narela 400kV D/c(Quad) lines formed after removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh/Gopalpur S/s and Extension of above LILO section from Maharani Bagh/Gopalpur upto Narela S/s

*****Gopalpur Substation is under implementation by DTL by LILO of Bawana – Maharani Bagh 400kV D/c(Quad) at Gopalpur substation.***

The Rajasthan SEZ Ph-II (8.1GW) Part G is under advance stage of implementation and 765/400kV Narela substation along with its connectivity to Maharani Bagh is expected to be commissioned by Dep’24.

Earlier, in the 39th Meeting of the Standing Committee on Power System Planning of Northern Region held on 29-30th May’17, following intra state transmission scheme was agreed for implementation by DTL

- Establishment of 4x500MVA, 400/220kV GIS Substation at Gopalpur along with 125 MVAR bus reactor
- LILO of Maharanibagh–Bawana 400 kV D/C line at Gopalpur 400/220kV substation on multicircuit towers

Subsequently, a meeting was held in CEA on 29.5.2019, wherein, DTL informed that works for Gopalpur 400 KV sub-station are presently under tendering process and the substation is likely to be commissioned by 2021. In view of above system studies were carried out considering Gopalpur substation and a new 765/400kV Narela substation (ISTS) was planned as part of transmission scheme for Rajasthan SEZ Ph-II (8.1GW) transmission scheme considering connectivity with 400/220kV Gopalpur substation. The scheme was agreed in 5th NRSCT meeting held on 13.09.19.

Further, in the 2nd Meeting of Northern Region Power Committee (Transmission Planning) (NRPCTP) held on 01.09.20, Chairperson, CEA enquired about the status of Gopalpur S/s. DTL replied that the Gopalpur S/s is at tendering stage and will be commissioned by 2023.

As per latest information provided by DTL, the scheme is taken up for board approval with implementation schedule of 30 months from award. The issue was also highlighted in the 209th OCC meeting held on 19.07.2023 and 69th NRPC meeting held on 27.09.23. In the meeting it was deliberated that Gopalpur Sub-station is not even yet awarded by DTL. In absence of Gopalpur Substation, loadings on 400/220kV ICTs at Maharani Bagh may become critical in future in solar maximized scenario which may impact RE evacuation. In the meeting, it was also suggested that in the meantime (till availability of Gopalpur S/s), DTL may explore load segregation at Maharani Bagh Sub-station so as to contain ICT loadings in solar maximized scenario for 2024-25 & beyond till availability of Gopalpur S/s. Subsequently NRPC also convened a meeting on above issue on 05/10/23.

CTU requested DTL to update the status of implementation of 400/220kV Gopalpur S/s and share how they will manage the load to contain Maharani Bagh ICT loadings within permissible limit. DTL stated that approval for 400/220kV Gopalpur and associated transmission system is going in next board meeting and award is expected by Mar'25 with 30 months schedule (Schedule: Sep'27). CEA enquired plan for load management by DTL in absence of Gopalpur S/s. DTL further stated that load shall be segregated/radialized among Maharani Bagh, Tughlakabad & Harsh Vihar S/s in future in such a way that Maharani Bagh ICT loadings does not not breach N-1 limits. This feeder segregation/radialisation is already followed by DTL at many other 400kV sub stations to address the high loading & same will be implemented in case of Maharani Bagh also to address the loading whenever required.

It was also requested that DTL may expedite the implementation of 400/220kV Gopalpur S/s and till its availability, ICT loadings at Maharani Bagh S/s may be closely observed to not breach N-1 limits through measures in intra state transmission network i.e. load segregation/radialization of feeders.

Based on above , it was agreed that availability of transmission system i.e. 400kV Gopalpur S/s along with LILO of Maharanibagh–Bawana 400 kV D/C line at Gopalpur 400/220kV substation on multi circuit towers (thus forming Maharanibagh – Narela 400kV D/c(Quad) and Maharanibagh -Gopalpur-Narela 400kV D/c(Quad) lines) shall not be the pre-requisite for effectiveness of LTA (now Connectivity under GNA) to the RE grantees utilising Rajasthan Ph-II (8.1GW) transmission System and based on commissioning of other transmission elements, Connectivity/GNA shall be made effective**.**

B.2 Augmentation of 1x500 MVA, 400/220 kV ICT at Bhadla-II PS (4th ICT at Section-1A)

It was stated that in the 8th CMETS-NR meeting held on 30.06.22 and 56th NRPC meeting held on 29.07.22, agenda for Implementation of "N-1" contingency at RE pooling substations in NR was deliberated. In the meeting Augmentation with 400/220kV, 1x500MVA Transformer at Bhadla-II PS (4th ICT at Section-1A) was agreed to be taken up for implementation with 15 months from the date of allocation of project or evacuation requirement beyond 1000 MW at 220kV level of Bhadla-2(Section-1A) whichever is later

Further agenda for implementation of "N-1" contingency at RE pooling substations in NR was again deliberated in the 73rd NRPC meeting held on 21.05.24. In the meeting it was mentioned that under Connectivity Regulations 2009, Connectivity & LTA were different products. However, subsequent to notification of GNA regulations, connectivity has become a single merged product which is granted along with equal quantum of GNA. Furthermore, implementation timeframe for augmentation of ICT was kept as 18 months instead of 15 months due to practical issues.

In the NRPC meeting, it was deliberated that augmentation with 400/220kV, 1x500MVA Transformer at Bhadla-2 PS (4th ICT at Section-1A) shall be required to meet N-1 criteria. At present, trigger conditions for taking up implementation of above ICT are yet to be happen, same was put up for approval for taking up implementation based on meeting the requirement stipulated as under:

- Augmentation with 400/220kV, 1x500MVA Transformer at Bhadla-2 PS (4th ICT at Section-1A)

  Implementation Timeframe: 18 months from the date of allocation of project (immediate evacuation requirement beyond 1000 MW at 220kV level of Bhadla-2(Section-1A))

In the 73rd NRPC meeting, proposal was technically agreed for taking up implementation of 400/220kV, 1x500MVA Transformer at Bhadla-2 PS (4th ICT at Section-1A) based on the trigger point requirement i.e. 18 months from the date of allocation of project (immediate evacuation requirement beyond 1000 MW at 220kV level of Bhadla-2(Section-1A)) or N-1 implication, whichever condition arrives earlier. It was agreed that above ICT will be taken up for implementation for which CTUIL will take up for the approval of above ICT based on its delegation of power without approaching the NRPC forum.

At present Bhadla-II PS (Sec-1A) has 400/220 kV transformation capacity of 1500 MVA(3x500MVA) whereas connectivity granted is 1520 MW and all the connectivity grantees have already completed the transition requirement of connectivity from Connectivity regulations 2009 to GNA Regulations 2022. As per manual on Transmission planning criteria 2023 "N-1' reliability criteria may be considered for ICTs at the ISTS / STU pooling stations for renewable energy based generation of more than 1000 MW after considering the capacity factor of renewable generating stations". Therefore, with 1520 MW of Connectivity at 220kV level of Bhadla-II PS(Sec-1A), implementation of 4th, 400/220 kV ICT needs to be taken up to meet N-1 reliability compliance at 220kV level.

In the meeting, NRLDC stated that considering 1520 MW of Connectivity at 220kV level of Bhadla-II PS(Sec-1A), 400/220 kV ICT augmentation is required for N-1 reliability compliance. In view of above, following ICT augmentation scheme was agreed in ISTS:

- Augmentation of 1x500 MVA, 400/220 kV ICT at Bhadla-II PS (4th ICT at Section-1A) along with associated transformer bays

  Implementation Schedule: 18 months from allocation

B.3 Augmentation of 1x500 MVA, 400/220 kV ICT at Fatehgarh-III PS (6th ICT in Section-2)

It was stated that at present, total connectivity granted at Fatehgarh-III PS(Sec-II) is 5520 MW, out of this, 2070 MW is granted at 220 kV level along with 50MW additional margin (at 220kV level) and 3450 MW is granted at 400 kV level. Establishment of Fatehgarh-III PS(Sec-II) and associated transmission system is currently under implementation as part of "Transmission system for evacuation of power from REZ in Rajasthan (20GW) under Phase-III Part E1 & E2" and expected to be commissioned progressively from Mar'25.

The above scheme involves establishment of 765/400/220 kV Fatehgarh-III PS(Sec-II) along with transformation capacity of 6x1500 MVA 765/400 kV ICTs & 5x500 MVA 400/220 kV ICTs. As per manual on Transmission planning criteria 2023 "N-1' reliability criteria may be considered for ICTs at the ISTS / STU pooling stations for renewable energy based generation of more than 1000 MW after considering the capacity factor of renewable generating stations". Considering the connectivity quantum of 2120 MW at 220 kV level (Connectivity granted: 2070MW, connectivity margin:50MW) at Fatehgarh-III PS(Sec-II), augmentation of 1x500 MVA ICT is required to be taken up to meet N-1 compliance.

In the meeting, NRLDC stated that considering the connectivity quantum (2120MW), augmentation with 400/220 kV, 315 MVA ICT may be considered in place of 500MVA ICT at Fatehgarh-III PS(Sec-II).

CTU stated that RE generator sometimes over inject (3-4%) in solar peak hours and this may lead to further increase in 400/220kV ICT loadings. Further with 315MVA ICT, total 400/220kV transformation capacity would be 2815MVA (2315 MVA in N-1) against connectivity quantum of 2120 MW at 220 kV level, very less margin available during solar peak hours after account for RE over injection, power factor & restricted over load capacity of ICTs in very high temperature condition of Rajasthan. Further, it was also stated that cost of 315MVA ICT is approx. 15-20% lower than 500MVA ICT. However, considering small difference in cost and homogeneity of all all 500 MVA transformer units at Fatehgarh-III PS, it was recommended that transformer augmentation may be carried out with 400/220 kV, 500MVA ICT (6th ICT in Sec-2) at Fatehgarh-III PS(Sec-II). CEA endorsed the same.

Accordingly, the following ICT augmentation scheme is agreed at Fatehgarh-III PS(Sec-II) PS in ISTS:

- Augmentation of 400/220 kV, 1x500 MVA (6th ICT in Sec-2) ICT at Fatehgarh-III PS(Sec-II) along with associated transformer bays

  Implementation Schedule: 18 months from allocation

B.4 Augmentation of 1x500 MVA(6th) , 400/220 kV ICT at Bikaner-III PS

It was stated that at present, total connectivity granted at Bikaner-III PS is 4667 MW, out of this, 2267 MW is granted at 220 kV level and 2400 MW is granted at 400 kV level. Establishment of Bikaner-III PS and associated transmission system is currently under implementation as part of "Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1) (Bikaner Complex)" and is expected to be commissioned progressively from Dec'25.

The above scheme involves establishment of 765/400/220 kV Bikaner-III PS along with transformation capacity of 6x1500 MVA 765/400 kV ICTs & 5x500 MVA 400/220 kV ICTs. As per manual on Transmission planning criteria 2023 "N-1' reliability criteria may be considered for ICTs at the ISTS / STU pooling stations for renewable energy based generation of more than 1000 MW after considering the capacity factor of renewable generating stations. Considering the connectivity granted of 2267 MW at 220 kV level at Bikaner-III PS, augmentation of 1x500 MVA ICT is required to be taken up to meet N-1 compliance.

Accordingly, the following ICT augmentation scheme is agreed at Bikaner-III PS in ISTS:

- Augmentation of 400/220 kV, 1x500 MVA (6th) ICT at Bikaner-III PS along with associated transformer bays

  Implementation Schedule: 18 months from allocation

B.5 Transmission System for connectivity of Pumped Storage Projects in Sonbhadra District in Uttar Pradesh:

It was stated that in the 31st CMETS-NR meeting held on 27.06.2, connectivity applications of cumulative quantum of 5152 MW from two developers i.e. M/s Greenko (6 nos. of applications with cumulative quantum of 4032 MW) & M/s Avaada WB (1 application of 1120MW) near Robertganj area in Sonbhadra district, Uttar Pradesh was discussed. As per the schedule indicated in the applications, these PSP projects are expected to be commissioned progressively from Nov'26 upto Mar'28. Details of applications are as under:

S.No	Applications	Connectivity (MW)	Start Date of Connectivity (As per Application)	Maximum Injection (MW)- As per application	Maximum Drawl (MW) – As per application
1	Greenko UP01 IREP Private Limited	672	16.11.2026	610	672
		672	01.02.2027	610	672
		672	16.04.2027	610	672
		672	16.04.2027	610	672
		672	01.07.2027	610	672
		672	16.12.2027	610	672
2	Avaada Waterbattery Private Limited	1120	31.03.2028	900	1120
	Total (MW)	**5152**		**4560**	**5152**

In the meeting, it was informed that to deliberate on planning of transmission System for evacuation of Power from Pumped Storage Plants, a meeting under Chairmanship of Chairperson, CEA was held on 28.05.2024. In the above meeting, it was decided that CTUIL while granting Connectivity to PSPs shall mention that PSPs shall not operate in generating mode during high RE generation period and if required, PSPs may inject power during high RE generation period based on margin available in the system. However, detail analysis must be done based on the combination of sources at that node.

M/s Greenko vide letter dated 21.05.24 informed that In the initial phases of PSP development, it is envisaged as a Stand-alone storage project, which means the PSP project will draw power from renewable generation sources or conventional generation sources located at different location(s) in the grid for charging the pump storage plant and thereafter power will be injected into the grid for supplying power to the different beneficiaries connected to different location(s) in the grid. During PSP operation, typically during peak solar time (mainly between 11AM to 2 PM), the PSP project shall be able to draw power to the extent of 4032 MW (6X672MW) corresponding to pumping capacities of all six units. While, during generation hours which will be typically during non-solar or low solar hours, the maximum generation possible shall be to the extent of 3660 MW (6x610MW). For evolving the transmission scheme, the above-mentioned maximum values of drawl and Generation may be considered. Same was also discussed in 31st CMETS-NR meeting.

CEA in the above CMETS-NR meeting confirmed that the Sonbhadra area may be considered as a potential zone for pumped storage projects. Accordingly, the transmission system for connectivity of M/s Greenko & M/s Avaada at Roberstganj PS shall be considered as common transmission system.

CEA also informed that TOR has been issued to M/s Greenko Sonbhadra PSP (3660 MW) and presently under S&I. M/s Avaada in the meeting informed that the TOR of their project is approved and currently in pre DPR stage & they shall be submitting it to CEA for pre DPR chapter approvals.

M/s Avaada & Greenko clarified in the meeting that they shall operate PSP units in synchronous condenser mode as per the reactive power requirement of the grid. Further, M/s Greenko and M/s Avaada shall also keep future provision of Bus Reactor so that in case of future requirement the same can be installed by the applicant.

Accordingly, in the above CMETS-NR meeting, connectivity applications of M/s Avaada & Greenko were agreed for grant, and it was deliberated that the transmission scheme for Pump Storage plants in Sonbhadra district is presently under discussion. The scheme is currently tentative, which shall be finalized in subsequent CMETS-NR and other region (ER) CMETS meeting among CEA, CTUIL, Grid India & Stakeholders. The detailed transmission system shall be informed upon finalization and approval of the scheme.

Further, NR-WR inter regional scheme to relieve loading of 765 kV Vindhyachal – Varanasi D/c line shall also be required for connectivity of Pumped Storage Projects in Sonbhadra district.

In view of the above, studies were carried out considering PSPs in drawl mode in Solar Max scenarios & injection mode in Peak load scenarios (evening peak). All India Study files for various scenarios (solar maximized, evening peak and night off peak) were circulated to NR stakeholders on 12.09.24.

Subsequently, M/s Greenko had submitted requisite Connectivity BGs only for three application (3x672MW), out of 6 nos. M/s Avaada also submitted requisite Connectivity BGs for their 1120MW PSP project, therefore transmission scheme needs to be revised in view of reduction in connectivity quantum by M/s Greenko.

M/s Greenko vide letter 10.09.24 requested that as they are moving forward with three applications for their project, there may be a change in the transmission scheme from the originally planned scheme involving six Grid Connectivity applications. During CTU's review of this change in the transmission system, it may be necessary to consider including the bay implementation for terminating the dedicated transmission line at Robertsganj substation under the ISTS scope

In view of schedule of generation projects and for optimal utilization of transmission scheme, comprehensive tr. scheme is planned considering M/s Greenko and M/s Avaada PSP evacuation requirement (Maximum injection 3136MW, Maximum Drawl : 2730MW) including future requirement. However, in view of transmission being lumped elements, the planned scheme can cater upto 4 GW PSP connectivity quantum.

In the meeting, Grid-India stated to review the reactive compensation of 765kV Robertsganj PS – Prayagraj S/s D/c line. CTU stated that with proposed line reactors (240MVAr line reactor on both ends), Reactive compensation is bit on higher sider (~90%), therefore 330MVAr line at one of the end may be considered for above line based on studies. Accordingly, 330MVAr line reactor is considered at Robertsganj PS end for 765kV Robertsganj PS – Prayagraj S/s D/c line. On the query of direction of power flow from Varanasi to Robertsganj and Gaya to Robertsganj in Solar peak hours, CTU stated that during solar peak hours, PSP may operate in pumping mode and power will flow from Varanasi to Robertsganj PS and Robertsganj PS to Gaya S/s.

Grid-India enquired about line rating of 400kV Greenko- Robertsganj D/c line. CTU stated that considering three applications of 2016MW (3x672MW) of M/s Greenko, rating of dedicated line shall be considered commensurate to power transfer requirement under N-1 contingency of PSP plant.

Grid-India enquired that line reactors may be reviewed w.r.t. line length of each sections formed after proposed LILO arrangements i.e. LILO of 765 kV Fatehpur-Varanasi S/c line at Prayagraj & LILO of 765 kV Fatehpur-Sasaram S/c line at Prayagraj. Grid-India stated that 765kV Fatehpur – Prayagraj section may get overcompensated with existing line reactor configuration. CTU stated that after proposed LILO arrangement, reactive compensation is about 89% on 765kV Fatehpur – Prayagraj section (140 kms). Grid-India enquired about feasibility of removing 330MVAr existing line reactor at Fatehpur end or its replacement with 240MVAr line reactor. CTU stated that considering prevailing high voltage condition in NR (in some off peak scenarios), removal or replacement of 330MVAr line reactor at Fatehpur end may worsen the high voltage problem, however line length is tentative and will be reviewed in Gati Shakti portal. In the case of reduction of line length considerably, decision may be taken on removal/replacement of 330MVAr line reactor at Fatehpur end of 765kV Fatehpur – Prayagraj section (140 kms). CTU further stated that on all other line sections formed after LILO arrangement at Prayagraj S/s, reactive compensation is in order.

Grid-India enquired that whether PSP developers indicated their machine type i.e. fixed speed or variable speed type. M/s Greenko informed that they have fixed speed machines. M/s Avaada informed that they will revert on the same. M/s Greenko requested that during finalization of the scheme, 400kV bays at Robertsganj end for termination of 400kV Greenko-Robertsganj D/c line may be considered in ISTS scope. CTU stated that as the scheme is still under approval and being revised in this meeting, same will be incorporated as part of comprehensive agreed transmission scheme and M/s Greenko shall have to fulfil Conn-BG2 requirement.

Grid-India also stated that 1200kV system may also be explored from above complex in future for evacuation of envisage PSP potential due to its proximal location form WR and ER region and firm power flow in most of the scenarios.

CTU further stated that based on agreed scheme in meeting, transmission system under applicant scope as well as common transmission system will be revised. It was also stated that M/s Avaada & Greenko shall operate PSP units in synchronous condenser mode (minimum one unit) as per the reactive power requirement of the grid.

CEA stated that both the projects are under advance stage, however DPR is yet to approve. On the query of CEA on additional generation considered at 400kV level, CTU informed that at present comprehensive scheme is planned for 4GW PSP potential and with connectivity quantum more than 4GW, adequacy of agreed transmission scheme shall be reviewed further. CEA and UPPTCL also agreed on the proposal

CTU enquired to UPPTCL about drawl requirement (220kV) from Robertsganj and Prayagraj S/s. UPPTCL stated that intra state transmission system (tentative) is already planned for evacuation of thermal power projects in the vicinity and in receipt of PSP application in future, new intra state schemes shall be planned. As part of above schemes, many new substations may be planned in intra state and can be utilized for drawl purpose of UP. Based on above, it was decided that provision of 220kV scope shall be deleted from future provisions at Robertsganj PS and Prayagraj PS, which was agreed by UPPTCL.

CTUIL in the meeting as well as their mail dated 20.09.24 requested POWERGRID to provide availability of OPGW on below lines so that provision of OPGW may be kept as part of present agreed scheme in case of non-availability of OPGW system

- 765 kV Fatehpur-Varanasi S/c line (proposed to be LILOed at Prayagraj S/s)
- 765 kV Fatehpur-Sasaram S/c line (proposed to be LILOed at Prayagraj S/s)
- 765 kV Varanasi- Gaya 2xS/c line (proposed to be LILOed at Robertsganj PS)

As per the inputs received from POWERGRID, OPGW is not available on the existing 765 kV Fatehpur-Varanasi S/c line & 765 kV Fatehpur-Sasaram S/c line which is proposed to be LILOed at Prayagraj S/s. Therefore, OPGW installation needs to be done to cater voice/data/tele-protection requirement for the proposed new Prayagraj S/s in matching time frame of comprehensive transmission scheme. Further as per above inputs, existing 765 kV Varanasi- Gaya 2xS/c line (proposed to be LILOed at Robertsganj PS), OPGW is available on the 2nd S/c of above line, therefore on other S/c line OPGW is not required as existing OPGW can be used for voice/data/tele-protection requirement for both the S/c lines as source and destination stations are same. No other comments were received.

Based on above deliberations, following ISTS transmission system for evacuation of power from Pumped Storage Projects in Sonbhadra District was agreed :

- Establishment of 4x1500 MVA 765/400 kV Robertsganj Pooling Station* near Robertsganj area in Sonbhadra distt. (Uttar Pradesh) along with 2x240 MVAr 765 kV & 2x125 MVAr 400 kV bus reactors

  Future provisions at Robertsganj PS (excl. scope for present scheme): Space for

  - 765/400kV ICTs along with bays- 2 nos.
  - 765 kV line bays along with switchable line reactors – 6 nos.
  - 765kV Bus Reactor along with bay: 1 no.
  - 400 kV line bays along with switchable line reactor –6 nos.
  - 400kV line bays : 4 nos.
  - 400 kV Bus Reactor along with bays: 1 no.
  - 400kV Sectionalization bay: 2 sets

  ***along with provision of 80MVAr spare reactor (Single phase), 110MVAr (Single phase) & 500MVA spare transformer unit (Single phase)**

- 400kV line bays (4 nos.) for connectivity of PSP generation project (M/s Avaada & M/s Greenko) at Robertsganj PS
- LILO of both circuits of 765 kV Varanasi- Gaya 2xS/c line at Robertsganj PS (~50 km) along with 240MVAr switchable line reactor at each ckt of Robertsganj PS end of 765 kV Robertsganj PS - Gaya 2xS/c line (after LILO)
- Establishment of 765 kV Prayagraj S/s* near Prayagraj(Uttar Pradesh) along with 2x330 MVAr 765 kV Bus reactors

  Future provisions at Prayagraj S/s (excl. scope for present scheme): Space for

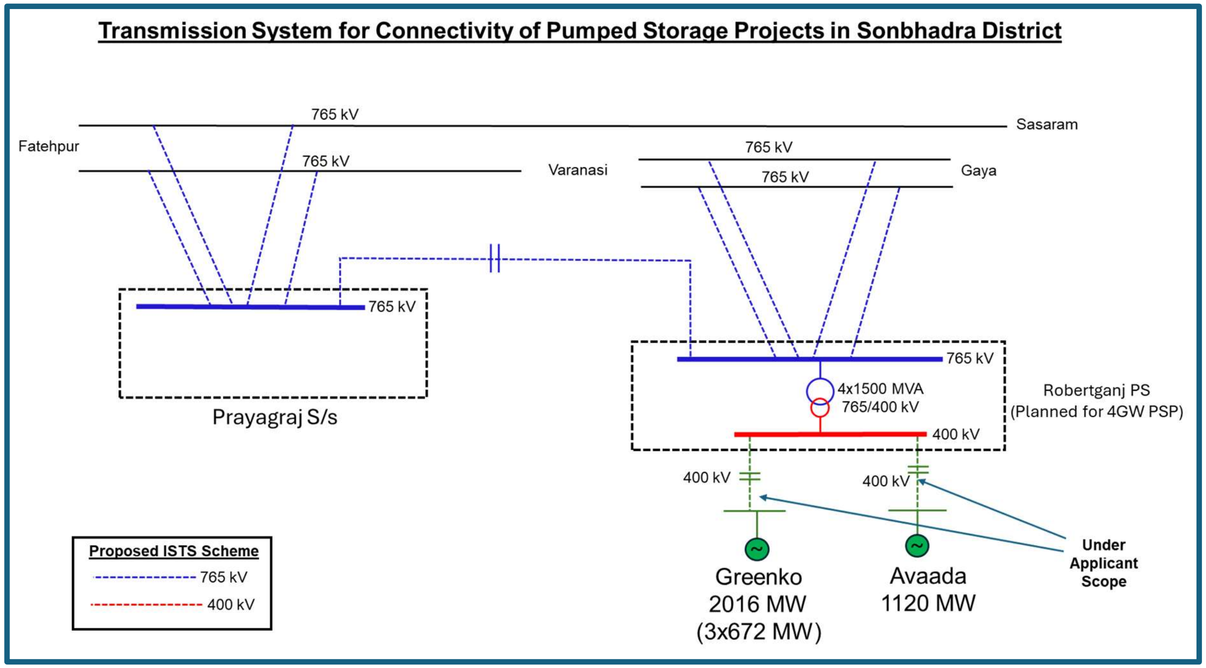
  - 765/400kV ICTs along with bays- 4 nos.
  - 765 kV line bays along with switchable line reactors – 4 nos.
  - 765kV Bus Reactor along with bay: 1 nos.
  - 400 kV line bays along with switchable line reactor –4 nos.
  - 400kv line bays : 2 nos.
  - 400 kV Bus Reactor along with bays: 2 no.
  - 400kV Sectionalization bay: 1 set

  ***along with provision of 110MVAr spare reactor (Single phase) & 500MVA spare transformer unit (Single phase)**

- LILO of 765 kV Fatehpur-Varanasi S/c line at Prayagraj (~30 km)

- LILO of 765 kV Fatehpur-Sasaram S/c line at Prayagraj (~30 km)
- Robertsganj PS – Prayagraj S/s 765 kV D/c line along with 330 MVAr line reactor at each circuit of Robertsganj end of Robertsganj PS – Prayagraj S/s 765 kV D/c line (~200 km)

From the studies, it was observed that considering drawl by PSPs in Peak RE hours (10AM to 2 PM), there is a requirement of additional inter-regional corridors towards WR which shall be discussed in subsequent section. However, it was gathered from Pump Storage project developers that they shall tie up with RE developers for drawing power in peak RE hours and shall be injecting power back in non RE peak hours(mainly during peak load period).



Further, Inter-regional (NR-WR) transmission System to relieve the loading of 765kV Vindhyachal-Varanasi D/c line was discussed and agreed in present meeting in subsequent section. This Inter regional transmission scheme is also required for evacuation of power from above Pumped Storage Projects in Sonbhadra District.

Transmission system for connectivity of M/s Greenko UP01 IREP Private Limited for its 6 nos. of connectivity applications ((App. No. 220000527-672 MW, 220000529-672 MW, 220000524-672 MW, 220000525-672 MW, 220000526-672 MW & 220000528-672 MW under GNA was agreed in 31st CMETS-NR meeting held on 27.06.24, which is as under:

<u>Details of Transmission system for Connectivity under GNA:</u>

A. <u>Associated Transmission System (ATS):</u> NIL

B. <u>Transmission System under applicant scope</u>

(i). Common Pooling station for Pumped Storage Projects of M/s Greenko UP01 IREP Private Limited (App. No. 220000527-672 MW, 220000529-672 MW, 220000524-672 MW, 220000525-672 MW, 220000526-672 MW & 220000528-672 MW) – Robertsganj Pooling Station 400 kV 2xD/c line(suitable to carry minimum 1400 MW per circuit at nominal voltage)*

**Minimum One unit to operate in Synchronous Condenser mode as per the requirement of the grid*

C. <u>Transmission system for Connectivity under GNA (Tentative):</u>

(i). Establishment of 5x1500 MVA 765/400 kV Robertsganj Pooling Station near Robertsganj area in Sonbhadra distt. Uttar Pradesh
(ii). LILO of both circuits of 765 kV Varanasi- Gaya 2xSc line at Robertsganj Pooling Station
(iii). Establishment of 765/400 kV 2x1500 MVA Prayagraj substation near Prayagraj(Uttar Pradesh)
(iv). Prayagraj – Sohawal 400 kV D/c line (Quad)
(v). LILO of 765 kV Fatehpur-Varanasi S/c line at Prayagraj S/s
(vi). LILO of 765 kV Fatehpur-Sasaram S/c line at Prayagraj S/s
(vii). Robertsganj PS – Prayagraj S/s 765 kV D/c line along with 240 MVAr line reactor for each circuit at Robertsganj end.

D. <u>Additional Inter regional (WR-NR)Transmission system for Connectivity under GNA (Tentative):</u>

(i). 765kV Vindhyachal Pool - Prayagraj D/c line along with 240MVAr line reactor (switchable) at Prayagraj end on each ckt of 765kV Vindhyachal Pool - Prayagraj D/c line
(ii). Bypassing of both ckts of 765kV Sasan – Vindhyachal Pool 2xS/c line at Vindhyachal Pool and connecting it with 765kV Vindhyachal Pool - Prayagraj D/c line, thus forming 765kV Sasan - Prayagraj D/c line

Start Date of Connectivity: 01.07.2027(interim). Final date shall be confirmed upon award of the system

Subsequently, as discussed above, M/s Greenko had submitted requisite Connectivity BGs only for three projects ((App No. 220000524-672 MW, 220000525-672 MW, 220000526-672 MW), out of 6 nos. of applications. In view of that revised transmission system for connectivity for three nos. of applications of M/s Greenko UP01 ((App No. 220000524-672 MW, 220000525-672 MW, 220000526-672 MW) is as under :

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

(ii). Common Pooling station for Pumped Storage Projects of M/s Greenko UP01 IREP Private Limited (App. No. 220000524-672 MW, 220000525-672 MW, 220000526-672 MW) – Robertsganj Pooling Station 400 kV D/c line(suitable to carry minimum 2016 MW per circuit at nominal voltage)*

***Minimum One unit to operate in Synchronous Condenser mode as per the requirement of the grid**

C. Transmission system for Connectivity under GNA (Tentative):

(i). Establishment of 4x1500 MVA 765/400 kV Robertsganj Pooling Station near Robertsganj area in Sonbhadra distt. (Uttar Pradesh) along with 2x240 MVAr 765 kV & 2x125 MVAr 400 kV Bus reactors
(ii). LILO of both circuits of 765 kV Varanasi- Gaya 2XS/c line at Robertsganj Pooling Station along with 240MVAr switchable line reactor at each ckt of Robertsganj PS end of 765 kV Robertsganj PS - Gaya 2xS/c line
(iii). Establishment of 765kV Prayagraj substation near Prayagraj(Uttar Pradesh) along with 2x330 MVAr 765 kV Bus reactors
(iv). LILO of 765 kV Fatehpur-Varanasi S/c line at Prayagraj S/s
(v). LILO of 765 kV Fatehpur-Sasaram S/c line at Prayagraj S/s
(vi). Robertsganj PS – Prayagraj 765 kV D/c line along with 330 MVAr line reactor at each circuit of Robertsganj end of Robertsganj PS – Prayagraj 765 kV D/c line

D. Additional Inter regional (WR-NR) Transmission system for Connectivity under GNA (Tentative):

(i). 765kV Vindhyachal Pool - Prayagraj D/c line along with 330MVAr line reactor (switchable) at Prayagraj end on each ckt of 765kV Vindhyachal Pool - Prayagraj D/c line
(ii). Bypassing of both ckts of 765kV Sasan – Vindhyachal Pool 2xS/c line at Vindhyachal Pool and connecting it with 765kV Vindhyachal Pool - Prayagraj D/c line, thus forming 765kV Sasan - Prayagraj D/c line

Start Date of Connectivity: 30.06.2027(interim). Final date shall be confirmed upon award of the system

Further, Transmission system for connectivity of M/s Avaada Waterbattery Private Limited under GNA, was also agreed in 31st CMETS-NR meeting held on 27.06.24, which is as under:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

(i). M/s Avaada Waterbattery Private Limited Pumped Storage Project– Robertsganj Pooling Station 400 kV D/c line (Quad moose or equivalent conductor suitable to carry 1716 MW per circuit at nominal voltage)

C. <u>Transmission system for Connectivity under GNA: (Tentative)</u>

(i). Establishment of 5x1500 MVA 765/400 kV Robertsganj Pooling Station near Robertsganj area in Sonbhadra distt. Uttar Pradesh
(ii). LILO of both circuits of 765 kV Varanasi- Gaya 2xSc line at Robertsganj Pooling Station
(iii). Establishment of 765/400 kV 2x1500 MVA Prayagraj substation near Prayagraj(Uttar Pradesh)
(iv). Prayagraj – Sohawal 400 kV D/c line (Quad)
(v). LILO of 765 kV Fatehpur-Varanasi S/c line at Prayagraj S/s
(vi). LILO of 765 kV Fatehpur-Sasaram S/c line at Prayagraj S/s

D. <u>Additional Inter regional (WR-NR) Transmission system for Connectivity under GNA (Tentative):</u>

(i). 765kV Vindhyachal Pool - Prayagraj D/c line along with 240MVAr line reactor (switchable) at Prayagraj end on each ckt of 765kV Vindhyachal Pool - Prayagraj D/c line
(ii). Bypassing of both ckts of 765kV Sasan – Vindhyachal Pool 2xS/c line at Vindhyachal Pool and connecting it with 765kV Vindhyachal Pool - Prayagraj D/c line, thus forming 765kV Sasan - Prayagraj D/c line

Start Date of Connectivity: 31.12.2028(interim). Final date shall be confirmed upon award of the system

In view of that revised transmission system for connectivity application of M/s Avaada Waterbattery ((App No. 220000553-1120 MW) is as under:

<u>Details of Revised Transmission system for Connectivity under GNA:</u>

A. <u>Associated Transmission System (ATS):</u> NIL

B. <u>Transmission System under applicant scope</u>

(i). M/s Avaada Waterbattery Private Limited Pumped Storage Project– Robertsganj Pooling Station 400 kV D/c line (Quad moose or equivalent conductor suitable to carry minimum 1716 MW per circuit at nominal voltage)

C. <u>Transmission system for Connectivity under GNA (Tentative):</u>

(i). Establishment of 4x1500 MVA 765/400 kV Robertsganj Pooling Station near Robertsganj area in Sonbhadra distt. (Uttar Pradesh) along with 2x240 MVAr 765 kV & 2x125 MVAr 400 kV Bus reactors
(ii). LILO of both circuits of 765 kV Varanasi- Gaya 2xSc line at Robertsganj Pooling Station along with 240MVAr switchable line reactor at each ckt of Robertsganj PS end of 765 kV Robertsganj PS - Gaya 2xS/c line
(iii). Establishment of 765kV Prayagraj substation near Prayagraj(Uttar Pradesh) along with 2x330 MVAr 765 kV Bus reactors
(iv). LILO of 765 kV Fatehpur-Varanasi S/c line at Prayagraj S/s
(v). LILO of 765 kV Fatehpur-Sasaram S/c line at Prayagraj S/s

(vi). Robertsganj PS – Prayagraj 765 kV D/c line along with 330 MVAr line reactor at each circuit of Robertsganj end of Robertsganj PS – Prayagraj 765 kV D/c line

D. <u>Additional Inter regional (WR-NR) Transmission system for Connectivity under GNA (Tentative):</u>

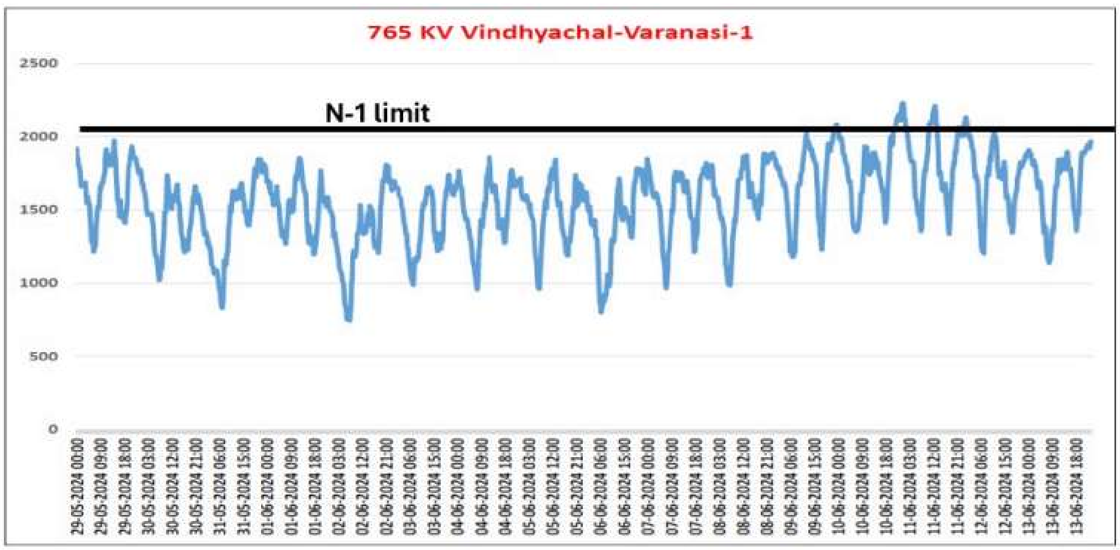
(i). 765kV Vindhyachal Pool - Prayagraj D/c line along with 330MVAr line reactor (switchable) at Prayagraj end on each ckt of 765kV Vindhyachal Pool - Prayagraj D/c line

(ii). Bypassing of both ckts of 765kV Sasan – Vindhyachal Pool 2xS/c line at Vindhyachal Pool and connecting it with 765kV Vindhyachal Pool - Prayagraj D/c line, thus forming 765kV Sasan - Prayagraj D/c line

Start Date of Connectivity: 31.12.2028(interim). Final date shall be confirmed upon award of the system

B.5 Inter-regional (NR-WR) Transmission System strengthening to relieve the loading of 765kV Vindhyachal-Varanasi D/c line

It was stated that in 220th NR-OCC meeting held on 19.06.24, it was deliberated that at the time of high demand in UP, it is being observed that the flow on WR-NR corridor is very high and issues related to high loading of 765 kV Vindhyachal – Varanasi D/C during high NR Import are being observed in real-time:



High loading, beyond N-1 limits of 765kV Vindhyachal-Varanasi D/C lines

Further, due to this high loading of 765kV Vindhyachal-Varanasi D/c, violation of WR-NR ATC and NR simultaneous import is also being observed in real-time. WR-NR ATC violations in real-time would lead to situation wherein NR states would not be able to draw further power from Western region and as a result, may need to resort to over drawl or load shedding in case internal generation in NR is not available

The issue was also highlighted by Grid-India in the meeting chaired by Addl. Secretary (MOP) on 29.05.24 & subsequent meeting in CEA regarding transmission constraints in Inter state and Intra state transmission system. It was deliberated that N-1 contingency of one ckt of 765kV Vindhayachal-Varanasi D/c line will over load the other ckt. In the meeting CTU stated that line loadings will be reviewed along with transmission scheme to be planned with PSP generation

In view of that studies were carried out and scheme was evolved to resolve critical loading of 765kV Vindhyachal-Varanasi D/c line in N-1 Contingency condition as well as for evacuation of PSP projects in UP.

In the meeting, UPPTCL stated that 765/400kV Fatehpur S/s is being implemented as part of Bhadla-III – Fatehpur HVDC transmission system. UPPTCL requested to explore feasibility of interconnection of Fatehpur S/s with Sasan/Vindhyachal PS to relieve high loading. CTU stated that as deliberated earlier, loading of 765kV Vindhyachal-Varanasi D/c line is already high and with envisaged PSP generation, loadings will further increase in 2027-28 time frame. CTU further stated that 765/400kV Fatehpur S/s will only be available in 2029 timeframe and about 300kms from Sasan S/s. Considering above it is not feasible to utilize Fatehpur S/s for WR-NR inter regional link. CTU informed that in planning of future ISTS transmission schemes, utilization of 765/400kV Fatehpur S/s (being implemented as part of HVDC scheme) shall be explored with additional envisaged PSP potential (beyond 4GW). Grid-India and CEA agreed on the proposal. It was also agreed that above strengthening scheme is required urgently to relive the loading of 765kV Vindhyachal-Varanasi D/c line. The NR-WR inter regional strengthening scheme was also discussed and agreed in 32nd CMETS-WR meeting held on 24.09.24

Based on above deliberations, following WR-NR Inter regional corridor is agreed in ISTS :

- 765kV Vindhyachal Pool - Prayagraj D/c line along with 330MVAr line reactor (switchable) at Prayagraj end on each ckt of 765kV Vindhyachal Pool - Prayagraj D/c line
- Bypassing of both ckts of 765kV Sasan – Vindhyachal Pool 2xS/c line at Vindhyachal Pool and connecting it with 765kV Vindhyachal Pool - Prayagraj D/c line, thus forming 765kV Sasan - Prayagraj D/c line

  Implementation Schedule: 24 months from allocation

B.6 Requirement of additional 1x500 MVA (5th), 400/220kV ICT at Maharani Bagh (PG) Substation

It was stated that at present 400/220kV Maharani Bagh has transformation capacity of 1630MVA (2x315MVA+2x500 MVA). From the loading pattern of Maharani Bagh ICTs as provided by Grid-India, it is observed that maximum loading on 400/220 kV ICTs is about 1397 MW (ICT-1-274MW, ICT-2-274MW, ICT-3-412 MW, ICT-4-437 MW) in past one year. From the analysis, it emerged that peak loading of ICTs are breaching N-1 limit (about 1300 MW) in summer season (May'24-Jul'24) for shorter duration of time. The loading pattern of Maharani Bagh ICTs for past one year is as below in fig:

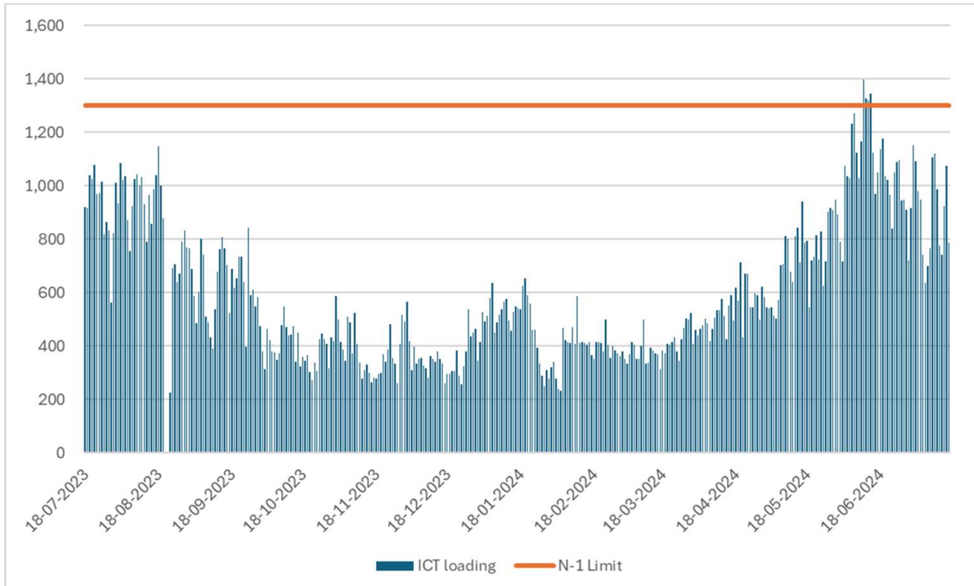


Fig : Loading of 400/220kV Maharani Bagh ICTs of Past one year (Source: Grid-India)

Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under phase-II Part G &G1 is under implementation (revised Schedule: Dec'24) comprising establishment of 765/400kv Narela S/s along with 765kV Khetri -Narela D/c line. The scheme also comprises removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh /Gopalpur S/s & Extension of above LILO section from Maharani Bagh/ Gopalpur upto Narela S/s to form Maharanibagh – Narela 400kV D/c(Quad) and Maharanibagh -Gopalpur-Narela 400kV D/c(Quad) lines.

Further as part of Transmission system for evacuation of power from REZ in Rajasthan (20GW) under Phase-III Part D, 765kV Sikar-II-Narela D/c line is under implementation (Sch. Aug'25). As discussed earlier, Gopalpur S/s is not even yet awarded by DTL and may be available by 2027-28, loadings on 400/220kV ICTs at Maharani Bagh may further increase (through interconnection of 400kV Narela-Maharani Bagh 2xD/c lines) with commissioning of Rajasthan Ph-II & Ph-III schemes which may impact RE evacuation.

In 2022-23 and 2023-24, Delhi had peak demand of about 7.5GW whereas in 2024-25 it witnessed exceptionally higher demand of 8.6GW (about 16% higher than 2023-24 peak demand). From the expected demand figures (as per 20th EPS), it is envisaged that Delhi demand may grow at about 5% (CAGR) and will rise to 9.5 GW by 2027.

POWERGRID vide letter dated 27.08.2024 to CTU also requested to review the requirement of additional ICTs at some of the substations (including Maharani Bagh S/s) of Northern region due to non-compliance of N-1 contingency. In the letter POWERGRID stated that, it has been observed that ICTs in few of the substations in Northern region are having high utilization and violating N-1 contingency criteria. POWERGRID highlighted that from the loading data at Maharani Bagh S/s (April-June 2024), it may be seen that ICTs Peak Utilization of 86% in June'24. POWERGRID vide mail dated 08.11.2023 confirmed space availability for Augmentation of 400/220 kV transformer (5^{th}, 500MVA) along with transformer bays at 400/220kV Maharani Bagh (PG) S/s.

In the meeting, NRLDC stated that at present 400/220kV ICTs at Maharani Bagh are not violating N-1 criteria, however ICTs breached N-1 limit (about 1300 MW) in the event of tower outage of 400kV Dadri-Harsh Vihar D/c (In May'24). In view of that option for replacement of 2x315MVA ICTs with 2X500MVA ICTs may also be explored. CTU stated that at present Maharani Bagh S/s has common 400kV common bus with two nos. 220kV sections having 2x500MVA ICTs in one section and 2X315MVA ICTs in other section connected through bus coupler. As per the information received from DTL, 220kV sectionalizer is proposed at Maharani Bagh S/s and both 220kV sections may operate separately in future. From the studies it is envisaged that with growing load & upcoming Narela transmission system, loading on 2x500MVA ICTs may increase more rapidly and may become N-1 non-compliant. Considering that ICT augmentation is required in the same section having 2x500MVA ICTs.

DTL stated they will require 400/220kV ICT augmentation at Maharani Bagh S/s as early as possible. DTL stated that for new 500MVA ICT augmentation, there may be space constraint for 220kV bay side. DTL also stated that 220kV Bus extension works at Maharani Bagh S/s is under tendering with 24 months schedule (from award). With above extension works 2 nos. 220kV bays will be available and may be utilized for above ICT augmentation, however schedule of above extension works may not align with ICT augmentation requirement and therefore site visit can be done to explore space availability of 220kV transformer bays or any interim arrangement at 220kV side for above proposed augmentation.

DTL also stated that additional load will be fed from Maharani Bagh S/s through proposed 2 nos. of 220/66kV & 3 nos. of 220/33kV transformers as part of 220kV bus extension scheme. CTU stated that DTL may ensure that with proposed additional load drawl from Maharani Bagh S/s, ICT loading remain balanced (in both 220kV sections) & within limit even with after proposed 400/220kV ICT augmentation. DTL agreed for the same.

POWERGRID confirmed that 315MVA ICTs only completed 17 years of life and working in healthy condition. POWERGRID stated that Prima facia space is available for 220kV transformer bay at DTL switchyard.

CEA enquired about incremental Maharani Bagh ICT loading without Gopalpur and associated transmission system. CTU stated that considering Rajasthan Ph-II & Ph-III transmission system and without Gopalpur transmission scheme, ICT loadings shall further increase to about 300MW-350MW (on 1630MVA) in solar maximized scenario in next 2.5-3 years and may become N-1 non-compliant. CEA agreed for augmentation of Maharani Bagh ICT augmentation. CEA enquired about short circuit level of Maharani Bagh S/s. CTU replied that short circuit level of 400kV Maharani Bagh S/s is 44kA (designed capacity: 40kA) and with proposed ICT augmentation, short circuit capacity will increase to 0.5kA. CEA suggested that 400kV bus splitting arrangement may be explored in consultation with POWERGRID/DTL with balanced load on each section.

Considering growing demand of Delhi (incl. additional load to be fed by Maharani Bagh S/s in 2027 timeframe) and delay in implementation of 400/220kV Gopalpur S/s by DTL, following ICT augmentation scheme is agreed in ISTS to meet N-1 compliance

- Augmentation of 400/220 kV ,1x500 MVA (5th) ICT at Maharani Bagh (PG) S/s along with associated transformer bays
- GIB duct at 400kV and 220kV level and cable at 220kV level

However, schedule of ICT will be finalized based on DTL confirmation (after site visit) on space availability of 220kV transformer bay as well as possibility of interim arrangement for 220kV side ICT bay space.

In case there is space constraint at 220kV side, schedule of above ICT augmentation will have to be aligned with 220kV bay extension works (24 months from award) being implemented by DTL as part of intra state scheme, for which award shall have to be confirmed by DTL for schedule alignment.

B.7 Requirement of additional 1x500 MVA (3rd), 400/220kV ICT at Lucknow (PG) Substation

It was stated that in 50th TCC & 74th NRPC Meeting held on 28-29th June 2024, NRLDC agenda for N-1 non-compliance of 400/220kV Lucknow (PG) ICT was deliberated. In the meeting it was informed that, for 400/220kV Lucknow (PG) ICT augmentation, confirmation from UPPTCL to be provided as Mohanlalganj is also commissioned near Lucknow. UPPTCL agreed to study requirement of ICT augmentation at 400/220kV Lucknow (PG) and revert before next NRPC meeting.

Further, in the Grid-India Quarterly operational feedback report (Q1 2024-25) of July 2024, issue of N-1 non compliance of ICTs at Lucknow (PG) S/s was highlighted. In the report it was mentioned that during high demand season, ICTs loading (750- 800MW) become N-1 non-compliant (N-1 loading limit of ICTs is 710MW) for sometimes in the month of Jun’24. The loading pattern of 400/220kV Lucknow (PG) ICTs for Q1 (2024-25) is as below fig.

400/220kV Lucknow(Pg) ICT loading for Q1 2024-2025

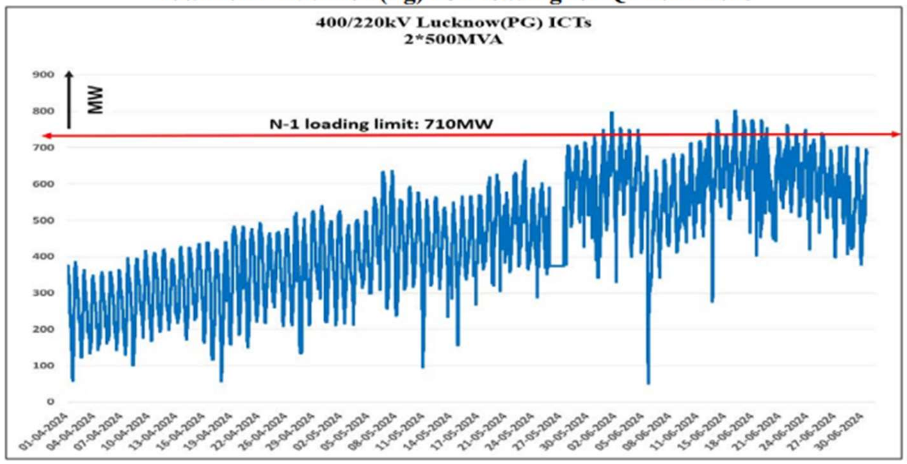


Fig : Loading of Lucknow (PG) ICTs of Q1 (2024-25) (Source: Grid-India Quarterly operational feedback report)

In 221st OCC meeting held 19.07.2024, POWERGRID agenda for increasing capacity of various ICT's incl. 400/220kV Lucknow(PG), in Northern region was deliberated. In the meeting POWERGRID apprised forum that 400/220kV Lucknow (PG) ICT are not complying N-1 contingency criteria under the peak loading scenario. Further POWERGRID mentioned that regarding addition of new ICT at Lucknow S/s, space requirement was identified and observed that space for new ICT is available at 400/220kV Lucknow (PG) (with provision of outdoor GIS bays). NRPC asked UPPTCL to give their inputs on the said matter to CTU in next 10 days.

Further, UPPTCL vide letter dated 30.08.2024 stated that, load flow study is done on various Substations in UP and accordingly additional 500 MVA ICT is required at Lucknow PG S/s. In the minutes of meeting (dated 05.09.24) of 222nd OCC-NR meeting held on 14.08.24 it is mentioned that that UPPTCL vide letter dated 30.08.2024 submitted its views/observation on the cited matter to NRPC, CEA and CTU & CTU to take up the issue in its next CMETS meeting.

POWERGRID vide mail 13.09.24 informed that 400/220kV ICT-3 at Lucknow S/s is feasible with optimized space. However, following constraints/ issues to be taken care while approval of scheme.

- As there is no space in the existing 400kV yard, hence, Proposed 400kV ICT bay shall be Outdoor GIS type
- Further Connectivity for 400kV Bus Extension to GIS Bays is to be achieved through GIB Duct having multiple road crossings in duct route from 400kV Bus to GIS bay
  220kV Side of ICT-3 bay shall be AIS type [Future Extension of 220kV Bays will be limited to 2 numbers future feeders].
  Further, High Voltage Test of 400kV GIS bay may require complete shut Down of bus

UPPTCL reiterated the requirement of Augmentation of 400/220 kV ,1x500 MVA (3rd) ICT at Lucknow (PG) S/s as load of Lucknow ICT will not be shared by other 400kV substations nearby. In the meeting, UPPTCL also stated that 2 nos. of 220kV line bays (future space) at Lucknow (PG) S/s are sufficient to meet their future drawl requirement from above S/s as they are already planning for new 400kV S/s at Barapanki and Sitapur in near vicinity to serve the future demand requirement. CEA & NRLDC agreed on the proposal.

In view of above, following ICT augmentation scheme was agreed in ISTS:

- Augmentation of 400/220 kV ,1x500 MVA (3rd) ICT at 400/220kV Lucknow (PG) S/s along with associated transformer bays (1 no. additional 400kV line bay is also being implemented for GIS diameter completion)
- 400kV ICT bay shall be Outdoor GIS type & 220kV ICT bay shall be AIS type
- Connectivity for 400kV Bus Extension to GIS Bays is to be achieved through GIB Duct having multiple road crossings in duct route from 400kV Bus to GIS bay

B.8 PSTCL proposal for LILO of both ckts of 220 kV Jalandhar(BBMB)- Jamalpur (BBMB) D/c line (owned by BBMB) at 220 kV Goraya S/s (PSTCL) Substation

It was stated that PSTCL vide mail & letter dated 14.08.24 to CTUIL sent their agenda for approval of LILO of both ckts of 220 kV Jalandhar (BBMB)- Jamalpur (BBMB) D/c line (line owned by BBMB) at 220 kV Goraya S/s for taking up in upcoming CMETS-NR meeting.

In the letter, it was stated that 220 kV Goraya S/s is currently connected to 220kV Jamsher (PSTCL) and 220 kV Jadala by LILO of GGSSTP Ropar- Jamsher line. Further, 220 kV Jamalpur is a very critical substation as it feeds the power supply to Ludhiana industry area. The Jamalpur S/s draws power mainly from Bhakhra and Ganguwal and supplies power to Dhandari Kalan-I and Dhandari Kalan-2. These two stations (Dhandari Kalan-I and Dhandari Kalan-2) take power from 400kV Ludhiana (PG) also. This power drawl becomes even more prominent during non-paddy scenario since the load in Ludhiana area (being industrial load) invariably remains the same but the hydel generation availability in Punjab is reduced.

Further, 220 kV substation Nurmehal is connected to 400 kV Nakodar S/s in radial mode and there is an urgent need to connect 220kV substation Nurmehal to another source to enhance reliability of the system for which new 220kV link between Nurmehal and Goraya has already been planned in the current MYT 2023-26. So, with the 220kV connectivity of Goraya with Nurmehal through D/c and LILO of Jalandhar-Jamalpur line at Goraya S/s, Jamalpur S/s will gain an additional source of power from 400kV Nakodar S/s via Nakodar – Nurmehal-Goraya route, thus enhancing the reliability of power supply.

PSTCL mentioned that to cater the ever-growing demand of the industry, LILO of one circuit of Dhandari Kalan I-Jamalpur at Sherpur (2x160MVA) is under progress and the work is likely to be completed by Oct'24. Apart from this a new 220kV station with a capacity of 2x160MVA at Giaspura is also planned with LILO of 220kV Ludhiana(PG) -Dhandari Kalan circuit. So, there is urgent need to cater the ever-growing upcoming load within the sources available with PSTCL as there are serious issues of ROW in Ludhiana city area. PSTCL intends to use capacity available at 400 KV Nakodar through BBMB Jamalpur during the period of low Hydel Generation of BBMB, which becomes all the more necessary as Himachal Pradesh shall be using some power at Tahliwal from the line connecting Jamalpur-Bhakra.

The proposed LILO of 220 kV Jalandhar—Jamalpur line at 220 kV Goraya S/s (which is presently bisecting the yard area of 220 kV Goraya) and erection of D/c line from 220 kV Nurmehal to 220 kV Goraya will resolve the above-mentioned issues as 220 kV Jamalpur will then draw substantial amount of power from 400 kV Nakodar via Nakodar—Nurmehal—Goraya route as well, thus giving relief to ICTs at 400 kV Ludhiana (in low hydro scenario) as well as increasing the reliability of the transmission system with now an additional source feeding Jamalpur.

PSTCL also stated that, a committee comprising of officers of PSTCL was also constituted to deliberate on the matter and visited 220kV Goraya S/s and following was emerged after discussion

- 220 kV BBMB Jalandhar—Jamalpur D/c line passing through the yard may be LILOed (both circuits) at 220 kV substation Goraya.
- substation Goraya may be connected with 220 kV substation Nurmehal with 220 kV D/c line.
- No additional 100 MVA, 220/66 kV transformer needs to be installed at 220 kV substation Goraya for N-1 contingency.
- Suitable space is to be provided between the existing 100 MVA 220/132 kV and 100 MVA 220/66 kV transformers at 220 kV substation Goraya by shifting the existing 100 MVA 220/132 kV for constructing firewall between them.
- SHB shall be required to extend suitably.

Further, if the LILO of 220 kV Jalandhar—Jamalpur D/c line at 220 kV Goraya is approved, the 4 Nos. 220 kV bays required at 220 kV substation Goraya for carrying out the above mentioned LILO shall be constructed by PSTCL and shall be the asset of PSTCL.

PSTCL in the letter stated that an agenda on the subject matter was put up initially in the 145th Power Sub Committee meeting dated 12.07.2022. However, after various deliberations in the meeting, BBMB opined that power flow wise there seems no requirement of LILO at Goraya Substation. In response, PSTCL provided its comments vide memo no. 586 dated 07.09.2022, asserting that with the LILO of 220kV Jalandhar-Jamalpur line D/c at Goraya and D/C line from 220kV Nurmehal to Goraya, Jamalpur will draw power from 400kV Nakodar via Nakodar—Nurmehal— Goraya corridor.

CTU vide mail 18.01.23 to stakeholders (CEA, NRLDC & BBMB) requested to provide observation on PSTCL proposal at the earliest. In reply, BBMB vide mail 08.02.23 intimated that Load Flow Sheets enclosed along with the subject cited proposal are not showing overall scenario of Jamalpur station regarding drawl of power after its interconnections with 400kV Nakodar via Nakodar-Nurmehal-Goraya route as well as power drawn by 220kV Goraya station after LILO of 220kV Jalandhar-Jamalpur D/c Line is not forthcoming on record. BBMB requested that comprehensive Load Flow Study with n-1 contingency along with Load Flow Study report from NRLDC is required to be taken up for further necessary action in the matter . As far as strengthening of 220kV Sub-station, Goraya is concerned there is no hindrance from 220kV Jalandhar-Jamalpur D/c Line which is passing through the 220kV Goraya Sub-station as ample space is available at 220kV Goraya Sub-station to run through 220kV Jalandhar-Jamalpur D/C Line on the Gantries without LILO of the said line.

Further CTU in its mail 03.05.24 to BBMB informed that PSTCL vide letter 04.09.23 requested to take-up the agenda for LILO of both ckts of 220kV Jalandhar – Jamalpur line D/c line (Owned by BBMB) at 220kV Goraya (PSTCL) S/s in CMETS-NR meeting, however as per the PSTCL letter BBMB opined that there is no urgent need of proposed LILO of both ckts of 220kV Jalandhar – Jamalpur line D/c line at Goraya Sub-Station. As per the letter, BBMB also declared that there will not be any significant advantage to Punjab on account of power flows with LILO of 220kV Jalandhar-Jamalpur line (D/C) at 220kV S/S Goraya. PSTCL also informed that revised studies were also discussed with BBMB in Sep/Oct'23 and Jan'24

In this regard, CTU requested BBMB to provide comments on PSTCL revised studies and their observations on proposed LILO so that proposal can be deliberated in CMETS-NR meeting followed by SCSTPPSP-NR meeting for stakeholder consultation. BBMB vide letter dated 09.05.24 intimated that .sld and .sav cases provided by PSTCL cannot be accessed by BBMB and it may be sent in pdf format, however with the available N-1 condition, load flow studies received following feeders are found to be overloaded i.e. Jamalpur-Dhandari Kalan-1, Dhandari Kalan-1-Ludhiana, Ganguwal- Jamalpur D/c etc. To discuss upon BBMB observations, PSTCL vide mail 12.07.24 requested to convene a meeting among CTU, CEA, BBMB, and PSTCL to deliberate on the subject matter and address pertinent issues.

PSTCL also stated in letter (14.08.24) that recently, the matter was also discussed in a meeting held on 02.08.2024 with BBMB which was chaired by Member (Power), BBMB and where Engineer-in-Chief/TS, PSTCL and Chief Engineer/TS, BBMB were also present. BBMB agreed in-principle to the proposal in the meeting and recommended that the agenda may be taken up in the upcoming CMETS and NRPC meetings. NRPC vide mail 16.08.24 to CTUIL also forwarded the PSTCL proposal for discussion.

From loading pattern of 220 kV Jalandhar—Jamalpur D/c line, it emerged that loading on above line is 50-100MW/ckt in summer season (Apr-Jul) and below 50MW rest of the year. Loading pattern of 220 kV Jalandhar(BBMB)—Jamalpur(BBMB) D/c line is as under :

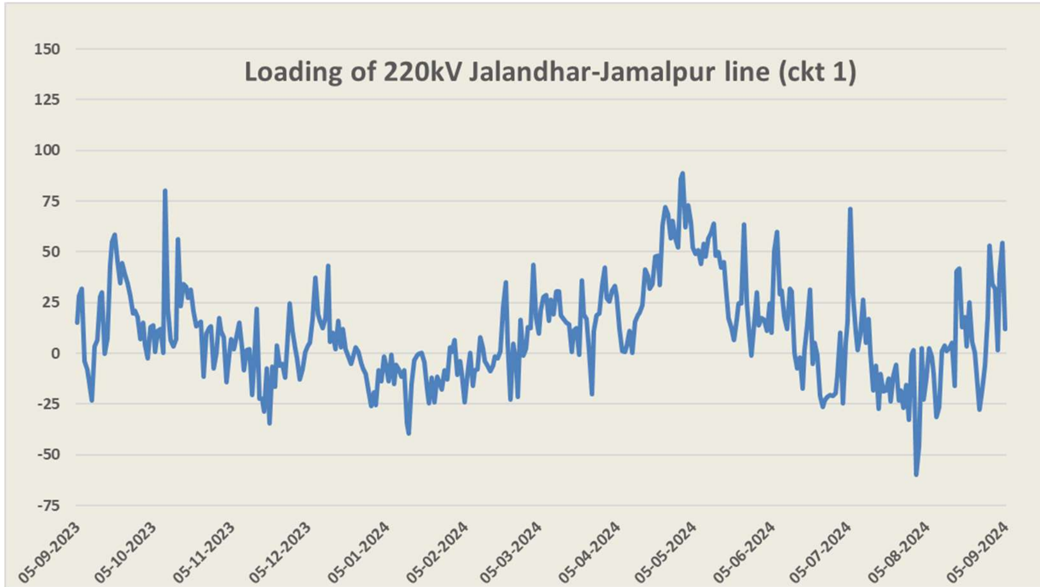


Fig : Loading of 220kV Jalandhar-Jamalpur D/c line (ckt1) (Source: Grid-India)

CTUIL also carried out studies on above proposal. As per the CTUIL studies, with proposed scheme i.e. LILO of 220 kV Jalandhar—Jamalpur D/c line at Goraya S/s (PSTCL), loadings are in order, whereas Ludhiana ICT (3x500+1x315MVA) loading is marginally increased (15-20MW/ICT).

BBMB stated that they already agreed in-principally on the proposal and from load flow studies there is not much loading increase on 220 kV Jalandhar (BBMB)- Jamalpur (BBMB) D/c line. PSTCL stated that LILO along with 4 nos. of 220kV line bays shall be implemented by PSTCL along with necessary communication requirement.

NRLDC stated that Jalandhar/Jamalpur area is mainly fed from 400kV Ludhiana, Ropar as well as Nakodar S/s. NRLDC requested PSTCL to update on 400/220kV Nakodar ICT augmentation. PSTCL stated that 400/220kV, 1x500MVA Nakodar ICT will be commissioned by 30.09.24 and total transformation capacity would be 1130MVA (2x315+1x500) by Dec'24. NRLDC stated that loading of 400/220kV Ludhiana ICT is also on higher side. PSTCL stated that with commissioning of 400/220kV, 1x500MVA ICT (2nd) at Dhanansu S/s which is expected in Dec'24, 400/220kV Ludhiana ICT may get relieved. NRLDC agreed for the above PSTCL proposal on LILO of both ckts of 220 kV Jalandhar (BBMB)- Jamalpur (BBMB) line D/c at 220 kV Goraya S/s (PSTCL) S/s.

PSTCL stated that they have sent another agenda on 20.09.24 (today) for replacement of 2 nos. 220/66kV, 100MVA ICTs with 160MVA at 220kV Jalandhar (BBMB) S/s for which BBMB also given in principal approval in Joint meeting with BBMB & PSTCL. PSTCL stated the 220/66kV ICTs at Jalandhar (BBMB) S/s are critically loaded and N-1 non-compliant in peak load scenario. PSTCL requested to take up the agenda in present CMETS-NR meeting for approval.

On the CTU enquiry about ownership of 220/66kV ICTs at Jalandhar (BBMB) S/s, PSTCL stated that ICTs are owned by BBMB, however 160MVA ICT will be implemented by PSTCL as part of intra state scheme. CTU stated that the ownership of ICTs will change from interstate (BBMB) to Intra state (PSTCL), therefore BBMB and PSTCL may mutually discuss on utilization of 100MVA ICT or Dismantle/Decapitalization of 100MVA ICT based on its useful life. PSTCL stated that modalities for Dismantle/Decapitalization of 100MVA ICTs will be discussed with BBMB in their next power sub-committee meeting

CTU stated that BBMB confirmation for replacement of 2 nos. 220/66kV, 100MVA ICTs with 160MVA ICTs at 220kV Jalandhar (BBMB) S/s along with treatment of existing 100MVA ICTs after replacement with 160MVA ICTs is required.

CTU stated that as agenda is received today only, they will seek comments from stakeholders through mail and based on their observations and subcommittee decision on existing 100MVA ICT treatment, agenda will be taken up in next CMETS-NR meeting.

Based on above deliberations, following intra state scheme was agreed to be implemented by PSTCL along with necessary communication infrastructure.

- LILO of both ckts of 220 kV Jalandhar (BBMB)- Jamalpur (BBMB) D/c line at 220 kV Goraya S/s (PSTCL)

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Annexure-I

Transmission system for Connectivity under GNA at Sirohi-II PS (tentative):

1. Establishment of 5x500 MVA, 400/220kV S/s at a suitable location near Sirohi(HVDC) PS (Sirohi-II)
2. Sirohi(HVDC) PS (Sirohi-II) – Sirohi PS 400 kV D/c line(Quad)
3. Sirohi(HVDC) PS (Sirohi-II) – Bhinmal(PG) 400 kV D/c line(Quad)
4. Establishment of 6000 MW, Sirohi(HVDC) terminal station (4x1500 MW) at a suitable location near Sirohi(HVDC) substation
5. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
6. HVDC line between Sirohi (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
7. Associated EHVAC system strengthening in WR/SR/ER

Annexure-II

Transmission system for Connectivity under GNA at Merta-III S/s (tentative):

1. Establishment of 400/220 kV, 5x500 MVA S/s at suitable location near Merta (Merta-III Substation) along with 2x125 MVAr bus reactor
2. Merta-III PS – Merta-II PS 400 kV D/c line(Quad)
3. Establishment of 400 kV Beawar (HVDC) S/s at a suitable location near Beawar
4. Merta-III PS – Beawar(HVDC) PS 400 kV D/c line (Quad)
5. Beawar(HVDC) PS – Beawar 400 kV D/c line(Quad)
6. Establishment of 6000 MW, Beawar (HVDC) terminal station (4x1500 MW) at a suitable location near Beawar (HVDC) substation
7. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
8. HVDC line between Beawar (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
9. Associated EHVAC system strengthening in WR/SR/ER

Annexure-III

Transmission system for Connectivity under GNA at Bikaner-V PS(Tentative)

1. Establishment of 765/400kV, 4x1500 MVA pooling station at suitable location near Bikaner (Bikaner-V PS) along with 2x125 MVAr & 2x240 MVAr bus reactor
2. 400/220kV, 5x500 MVA ICTs at Bikaner-V PS
3. LILO of both ckts of 400kV Bikaner-II PS- Khetri D/c line at Bikaner-V PS
4. 765kV Bhadla-IV PS – Bikaner-V PS D/c line
5. Establishment of 6000 MW, ± 800 kV Bikaner-V (HVDC) terminal station (4x1500 MW) at suitable location near Bikaner
6. Establishment of 6000 MW, ±800 kV Begunia (HVDC) terminal station (4x1500 MW) at Begunia (Distt. Khordha), Orissa (ER)
7. ±800 kV HVDC line between Bikaner-V (HVDC) & Begunia (HVDC) (with Dedicated Metallic Return)
8. Establishment of 765/400kV, 5x1500 MVA S/s substation station at Begunia along with 2x125 MVAr & 2x240 MVAr bus reactor
9. Begunia – Paradeep (ISTS) 765kV D/c line along with associated bays at both ends
10. Begunia – Gopalpur (ISTS) 765kV D/c line along with associated bays at both ends along with 240MVAr switchable line reactor for each circuit at Begunia end of Begunia –Gopalpur (ISTS) 765kV D/c line
11. Begunia – Khuntuni (OPTCL) 765kV D/c line

Annexure-IV

Transmission system for Connectivity under GNA at Ramgarh-II PS (Tentative)

1. Establishment of 2x1500 MVA, 765/400 kV Ramgarh-II Pooling Station at a suitable location near Ramgarh along with 2x125 MVAr & 2x240 MVAr bus reactor
2. Ramgarh-II PS – Bhadla-IV PS 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at Ramgarh-II PS end
3. Ramgarh PS – Ramgarh-II PS 400 kV D/c line (Quad)
4. LILO of both ckts of 765kV Ramgarh PS – Bhadla-III PS D/c line at Ramgarh-II PS along with 240 MVAr switchable line reactor for each circuit at Ramgarh-II PS end
5. Establishment of 6000 MW, Ramgarh(HVDC) terminal station (4x1500 MW) at a suitable location near Ramgarh-II substation
6. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
7. HVDC line between Ramgarh (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
8. Associated EHVAC system strengthening in WR/SR/ER
9. 2x[+]300/(-)150MVAr Synchronous condenser at 400kV level of Ramgarh-II PS

Additional system for connectivity at 220kV level only of Ramgarh-II PS

1. 400/220kV, 3x500 MVA ICTs at Ramgarh-II PS

<u>Annexure-V</u>

Transmission system for Connectivity under GNA at Barmer-III PS(Tentative)

For connectivity at 220kV or 400kV level of Barmer-III PS

1. Establishment of 400/220kV 5x500MVA Pooling Station at suitable location near Barmer (Barmer-III PS) along with 2x125 MVAr bus reactor
2. LILO of both ckts of 400kV Barmer-I PS – Barmer-II PS D/c line (Quad) at Barmer-III PS
3. Establishment of 6000 MW, Barmer-III(HVDC) terminal station (4x1500 MW) at a suitable location near Barmer-III PS
4. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
5. HVDC line between Barmer-III (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
6. Associated EHVAC system strengthening in WR/SR/ER
7. 2x[+]300/(-)150MVAr Synchronous condenser at 400kV level of Barmer-III PS

Annexure-VI

Transmission system for Connectivity under GNA at Fatehgarh -II PS(Tentative)

For connectivity at 400kV level of Fatehgarh-II PS

1. Augmentation with 765/400kV, 2x1500 MVA Transformer (6th & 7th) at Fatehgarh-II PS
2. Bhadla-II PS – Sikar-II 765kV D/c line(2nd)
3. Sikar-II – Khetri 765 kV D/c line
4. Sikar-II – Narela 765 kV D/c line along with 240 MVAr Switchable line reactor for each circuit at each end of Sikar-II – Narela 765 kV D/c line
5. Jhatikara – Dwarka 400kV D/c line (Twin HTLS*)
6. Augmentation of 1x1500 MVA ICT (3rd), 765/400kV ICT at Jhatikara Substation (Bamnoli/Dwarka section)
7. Sirohi – Mandsaur PS 765KV D/c line along with 240 MVAr switchable line reactor at Sirohi S/s end 330MVAr switchable line reactor at Mandsaur PS end for each circuit of Sirohi – Mandsaur PS 765KV D/c line
8. Mandsaur PS – Khandwa (New) 765kV D/c line along with 240MVAr switchable line reactor for each circuit at each end of Mandsaur PS – Khandwa (New) 765kV D/c line

* with minimum capacity of 2100 MVA/ckt at nominal voltage

Annexure-VII

Transmission system for Connectivity under GNA at Bhadla-IV PS(Tentative)

For connectivity at 220kV or 400kV level of Bhadla-IV PS

1. Establishment of 765/400kV, 4x1500 MVA, 400/220kV, 4x500 MVA ICTs pooling station at suitable location near Bhadla (Bhadla-IV PS) along with 2x125 MVAr & 2x240 MVAr bus reactor along with 2x125 MVAr & 2x240 MVAr bus reactor
2. Bhadla-IV PS – Bhadla-III PS 400kV D/c line (Quad)
3. 765kV Bhadla-IV PS – Bikaner-V PS D/c line
4. Establishment of 6000 MW, ± 800 kV Bhadla-IV (HVDC) terminal station (4x1500 MW) at suitable location near Bhadla
5. Establishment of 6000 MW, ±800 kV terminal station (4x1500 MW) at suitable location in WR/ER
6. ±800 kV HVDC line between Bhadla-IV (HVDC) & Suitable location in WR/ER (with Dedicated Metallic Return)
7. Establishment of 765/400kV, 2x1500 MVA S/s substation station at Suitable location in WR/ER along with 2x125 MVAr & 2x240 MVAr bus reactor
8. Associated transmission system in WR/ER for onwards dispersal of power

Additional system for connectivity at 220kV level only of Bhadla-IV PS

1. 400/220kV, 4x500 MVA ICTs at Bhadla-IV PS

Annexure-VIII

Transmission system for Connectivity under GNA at Pali PS (tentative):

1. Establishment of 400/220 kV, 5x500 MVA S/s at suitable location near Pali (Pali PS) along with 2x125 MVAr bus reactor
2. Pali PS – Beawar (HVDC) S/s 400 kV D/c line(Quad)
3. Establishment of 400 kV Beawar (HVDC) S/s at a suitable location near Beawar
4. Beawar(HVDC) PS – Beawar 400 kV D/c line(Quad)
5. Establishment of 6000 MW, Beawar (HVDC) terminal station (4x1500 MW) at a suitable location near Beawar (HVDC) substation
6. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
7. HVDC line between Beawar (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
8. Associated EHVAC system strengthening in WR/SR/ER

Annexure-IX

Transmission system for Connectivity under GNA at Jalore PS (tentative):

1. 5x500 MVA, 400/220kV ICTs at Jalore PS
2. Establishment of 400 kV Sirohi(HVDC) PS (Sirohi-II) at a suitable location near Sirohi
3. Establishment of 400/220 kV Jalore PS at a suitable location near Jalore
4. Sirohi(HVDC) PS (Sirohi-II) – Sirohi PS 400 kV D/c line(Quad)
5. Sirohi(HVDC) PS(Sirohi-II) – Bhinmal(PG) 400 kV D/c line(Quad)
6. Jalore PS – Merta-II PS 400 kV D/c line(Quad)
7. Jalore PS – Sirohi(HVDC) PS (Sirohi-II) 400 kV 2xD/c line(Quad)
8. Establishment of 6000 MW, Sirohi(HVDC) terminal station (4x1500 MW) at a suitable location near Sirohi(HVDC) substation
9. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
10. HVDC line between Sirohi (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
11. Associated EHVAC system strengthening in WR/SR/ER

Annexure-X

Transmission system for Connectivity under GNA at Bhadla-II PS

For connectivity at 400kV level of Bhadla-II PS

1. Augmentation with 765/400kV, 1x1500 MVA Transformer (5th) at Bhadla-II PS
2. Bhadla-II PS – Sikar-II 765kV D/c line(2nd)
3. Sikar-II – Khetri 765 kV D/c line
4. Sikar-II – Narela 765 kV D/c line along with 240 MVAr Switchable line reactor for each circuit at each end of Sikar-II – Narela 765 kV D/c line
5. Jhatikara – Dwarka 400kV D/c line (Twin HTLS*)
6. Augmentation of 1x1500 MVA ICT (3rd), 765/400kV ICT at Jhatikara Substation (Bamnoli/Dwarka section)
7. Sirohi – Mandsaur PS 765KV D/c line along with 240 MVAr switchable line reactor at Sirohi S/s end 330MVAr switchable line reactor at Mandsaur PS end for each circuit of Sirohi – Mandsaur PS 765KV D/c line
8. Mandsaur PS – Khandwa (New) 765kV D/c line along with 240MVAr switchable line reactor for each circuit at each end of Mandsaur PS – Khandwa (New) 765kV D/c line

   * with minimum capacity of 2100 MVA/ckt at nominal voltage

Annexure-XI

List of Participants of 34th Consultation meeting for Evolving Transmission Schemes in NR held on 20.09.2024

CEA

Shri Nitin Deswal	Deputy Director
Shri Kanhaiya Singh Kushwaha	Assistant Director

Grid India

Shri Priyam Jain	Ch. Manager
Shri Gaurav Malviya	Manager
Shri Gaurav dash	Dy. Manager
Shri Raj Kishan	Assistant Manager

CTU

Shri V Thiagarajan	Sr. GM
Shri Kashish Bhambhani	GM
Shri Sandeep Kumawat	DGM
Shri Tej Prakash Verma	DGM
Shri R Narendra Sathvik	Ch. Manager
Shri Yatin Sharma	Manager
Shri Madhusudan Meena	Engineer

POWERGRID

Shri Gunjan Agrawal	GM
Shri Rohit Kumar Jain	DGM

DTL

Shri Pramod Kumar	Dy. Manager

RVPNL

Dr. Om Prakash Mahela	XEN(PP&D)

PTCUL	
Shri Ashok Kumar	EE
Shri Neeraj Pathak	SE
Shri Amit Kumar Singh	SE
Shri Sanjeev Kumar Gupta	SE
Shri Ravi Shankar	SE
PSTCL	
Shri Sanjeev Sood	EIC/TS
HVPNL	
Shri Deepak Sarit	XEN SS
Shri Sanjay verma	SE
UPPTCL	
Shri Satyendra Kumar	SE
BBMB	
Shri Jagjeet Singh	
Shri R.K.Goyal	Dy. CE
Connectivity/GNA Applicants	
Shri Yogesh Sanklecha	ACME Solar Holdings Private Limited
Shri Lokesh Yadav	ACME Solar Holdings Private Limited
Shri Angshuman Rudra	Avaada Energy Private Limited
Smt. Shipra Arora	Avaada Energy Private Limited
Shri Ranjan Kumar	Ambuja Cements Limited, ACC Limited & Asian Fine Cements Pvt. Ltd.
Shri Rahul Malik	Amplus Centaur Solar Private Limited
Shri Devendra Sikarwar	Amplus Centaur Solar Private Limited
Shri Mahendra Singh Dabi	Adani Renewable Energy Holding Eighteen Limited
Shri Rajesh Kumar Gupta	Adani Renewable Energy Holding Eighteen Limited
Mr. Md Fahim Alam	Diyos Three Renewables Private Limited

Smt. Rasika Ghongade	Enren-III Energy Private Limited
Shri Saurabh Gupta	Enren-III Energy Private Limited
Shri Rohit Gera	Foxtrot Solar Renewables Energy Private Limited
Shri Sharul Khan	Foxtrot Solar Renewables Energy Private Limited
Smt. Priyanka Raj	Juniper Green Energy Private Limited
Shri Deepak Kumar Prajapat	Juniper Green Energy Private Limited
Shri Ravinder Kumar Rana	Prespa Solar Private Limited
Shri Mohit Jain	Renew Solar Power Private Limited
Smt. Aakanksha Naresh Bhisikar	Serentica Renewables India Private Limited
Shri Sirin Dutta Chowdhury	Serentica Renewables India Private Limited
Shri Shashank Gupta	Sunsure Solarpark RJ One Private Limited
Shri Rahul Choudhary	Sunbreeze Renewables Seven Private Limited
Shri Shatrughan Kushwaha	Sunbreeze Renewables Seven Private Limited
Shri Ajay Kumar	Sunbreeze Renewables Seven Private Limited
Shri Navneet Mairal	Sunbreeze Renewables Seven Private Limited
Shri Vikas Tiwary	Sindhari Solar Power Private Limited
Shri Sudhanshu Vishwakarma	Sindhari Solar Power Private Limited
Shri Abhishek Dhaka	Soltown Infra Private Limited
Shri Rahul Gupta	Soltown Infra Private Limited
Shri Shubham Verma	SAEL Industries Limited
Shri K A Vishwanath	TEQ Green Power XV Private Limited
Shri Pushpinder Hira	Greenko Rj01 Irep Private Limited